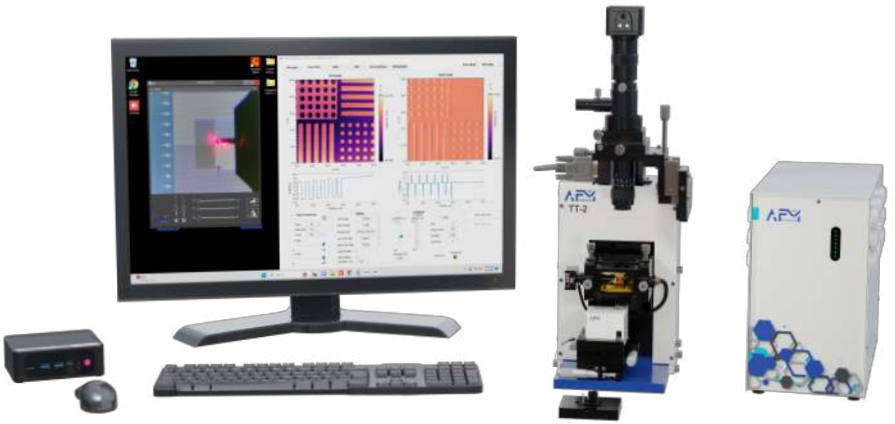
# TT-2 AFM

## Users Guide
## V 2.9



Part # 10-4172-10

# Definitions and Symbols

The following terms and symbols are used in this document and appear on the product where safety-related issues occur.

General Warning or Caution



The exclamation symbol may appear in warning and caution tables in this document. Symbol designates an area where personal injury or damage to the equipment is possible.

European Union CE Mark



The presence of the CE Mark on AFMWorkshop equipment means that it has been designed, tested, and certified as complying with all applicable European Union (CE) regulations and recommendations.

## Warnings/Cautions/Notes

The following are definitions of the warnings, cautions and notes that may be used in this manual to call attention to important information regarding personal safety, safety and preservation of the equipment, or important tips.

### WARNING

Situation has the potential to cause bodily harm or death.

### CAUTION

Situation has the potential to cause damage to property or equipment.

### NOTE

Additional information the user or operator should consider.

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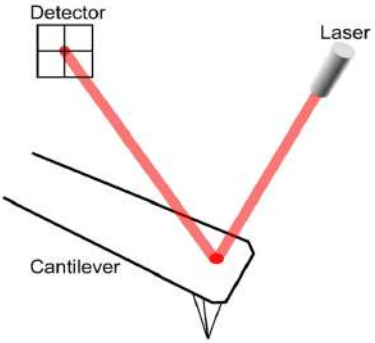
## NOTE

**Any and all mention of a 'laser' refers to the photoluminescent diode found in the TT-2 AFM. No true laser diodes are used in the AFM.**

# 1. Introduction to Atomic Force Microscopy

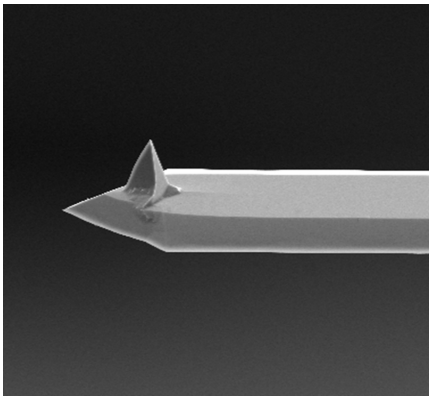
In an AFM (atomic force microscope), a probe is scanned over a surface and the motion of the probe is monitored to create a three-dimensional image of the surface. These unique instruments can measure high-resolution images in both ambient air and liquids, accurately determine the dimensions of surface features of only a few nanometers in size.

The three-dimensional motion of the sample (or probe) is generated by piezoelectric ceramics. These sensitive ceramics allow motions as small as a fraction of a nanometer. Typically, the sample (or probe) is moved in a raster pattern as the probe moves across the surface.




A light lever sensor is used for controlling the force of the probe on the surface while the sample is scanned. The light lever reflects a Photoluminescent Diode (PLD) beam off the surface of a cantilever into a photodetector. As the probe interacts with a surface, the cantilever deflects, and this motion is sensed by the photodetector.

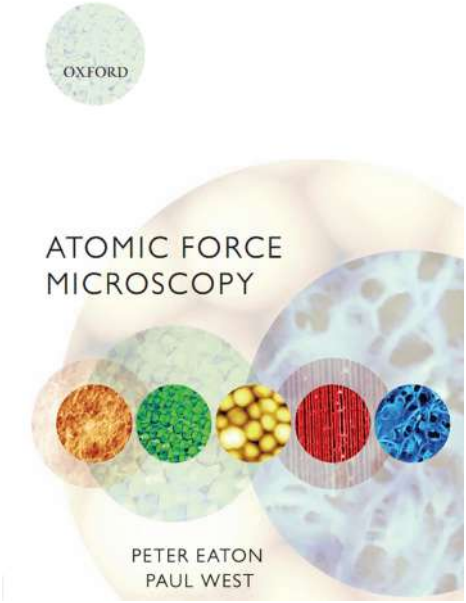
With this light lever, measurement and control of forces as small as a pico-newton are possible. With such small forces, very small probes may be used. With micro-machining methods, probes can have diameters of only a few nanometers.



The light lever can be made more sensitive by vibrating the cantilever with a small piezoelectric ceramic. When the vibrating probe interacts with the surface, the amplitude of vibration may be monitored and used to control the probe's force on the surface.

Modern atomic force microscopes include not only a probe and piezoelectric scanner, but additional hardware for bringing the probe rapidly into the proximity of a surface. A video optical microscope is very helpful for operating an AFM. The video microscope helps with aligning the light lever and probe approach, and for finding features for scanning.

For an in-depth description of AFM instrumentation, we recommend the book Atomic Force Microscopy by Peter Eaton and Paul West. This book provides a complete theoretical and practical explanation for the design and application of AFMs.



# 2. Introduction to the TT-2-AFM

When fully assembled, the TT-2 AFM comprises four subunits. They are the control computer, the Ebox, the stage, and the video optical microscope.

## 2.1 Stage

Samples are held and scanned on the AFM stage. On the upright inside the stage is a linear translator which moves the light lever force sensor in a vertical direction. The stage also includes the light lever, precision XY translators, and the piezoelectric scanner. Samples are held magnetically on a 1” diameter plate at the top of the piezoelectric scanner.

Optimal images are measured with the AFM stage if it is in a vibration- and acoustic-free environment. If necessary, a vibration and acoustic isolation system should be used. Appendix A provides more information on the best location for the AFM stage.

On the back cover of the stage is a modes connector. Signals required for implementing additional modes such as conductive AFM, STM and EFM are provided. Additional information on the cable configuration is provided in Appendix A.



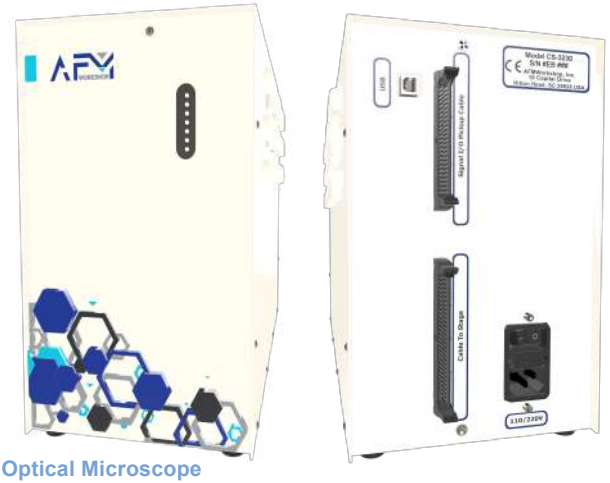

## 2.2 Computer

The control computer is a standard PC-type computer with a Microsoft Windows operating system. There are two programs required to operate the TT-2-AFM: the AFM control software and the software for the color video camera.



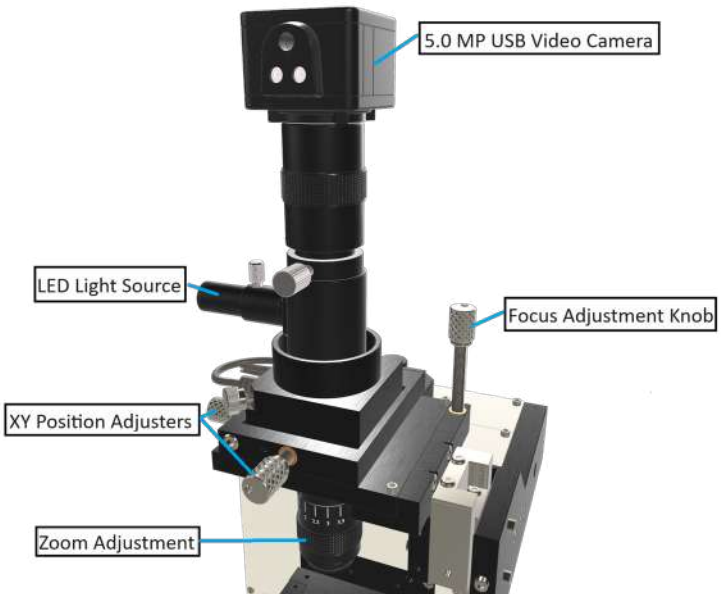
## 2.3 EBox

The EBox sends and receives signals from the computer through a single USB cable. Electronic signals are then sent to the stage through a 60-pin ribbon cable. Additionally, a grounding cable is connected between the stage and the EBox. All cables are connected at the rear of the EBox. In addition to the cable to the stage, there is a plug for an auxillary 50-pin ribbon cable that gives access to the EBox's internal electronic signals.



## 2.4 Video Optical Microscope

Aligning the light lever force sensor is greatly facilitated by the aid of the video optical microscope. The video microscope can help locate features on a surface for scanning. With the aid of the video microscope, tip approach can be executed much more rapidly. Images from the video microscope are displayed on the computer's video monitor..




# 3. Software

The TT-2 AFM includes three separate software modules in the AFM Installation files: AFM Workshop Acquisition Software, Video Microscope Software, and Gwyddion Image Analysis Software. This installation guide is targeted towards a computer running Windows OS.

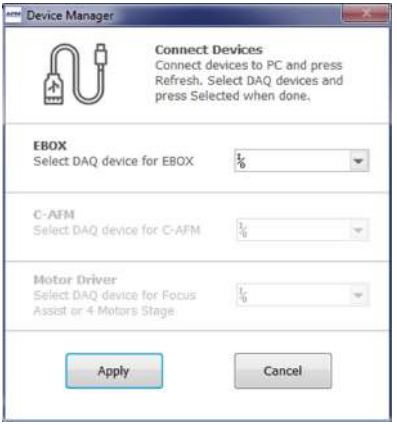
## 3.1 Installation Procedure

**NOTE: If the target computer has a previous version of LabView installed, it is imparative that it is completely removed prior to a new installation.**

Locate the provided AFM installation files for the Video Camera, Gwyddion, and AFMWorkshop Control softwares and follow the prompts. The user may be asked to input his/her scanner size (legacy software), so that the setup procedure will copy the files that correlate with that scanner size onto the desktop. Reboot the system when directed. All instructions can be found in the readme file.

**NOTE: When installing fresh software, a Windows Security pop-up may appear when attempting to run the software. If this occurs, please allow the software to run. This message will not reoccur.**

Upon initializing the AFMControl Software for the first time, the user will be asked to confirm the serial number of the National Instruments DAQ card in the EBox. Typically, there will only be one card option available. This will pop-up whenever the EBox is changed or new software is reinstalled.




### 3.1.1 Gwyddion Image Analysis Software

Gwyddion is an open-source SPM analysis software and can be downloaded for free on the internet at https://gwyddion.net/. The functions of Gwyddion Image Analysis software are:

- Processing routines such as leveling, deglitching, filtering, and smoothing which alter the images and data

- Display functions which change how the data is viewed. This includes 2D and 3D displays, as well as light shading and color mapping features.

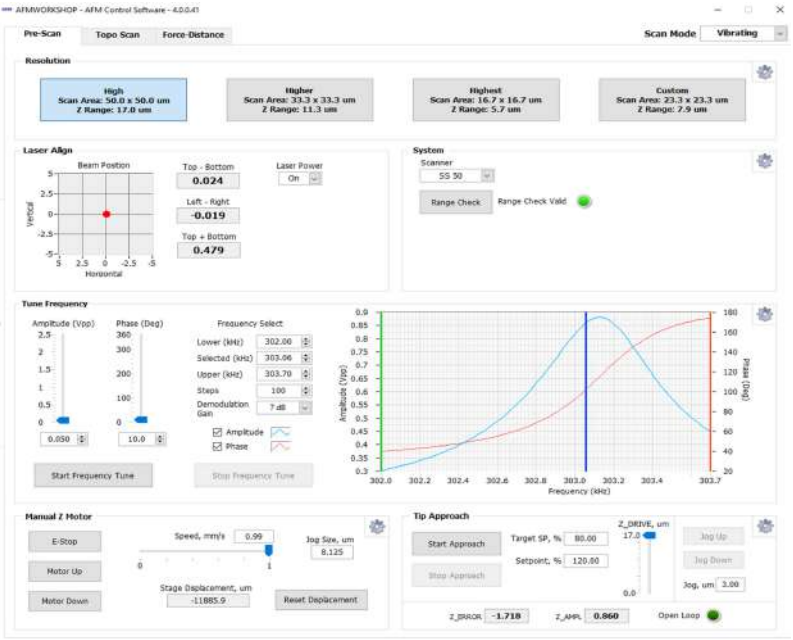
- Analysis options used for obtaining quantitative measurements from AFM images, such as line profiles, particle analysis, roughness measurements, and histogram analysis.

## 3.2 AFM Control Software

Once launched, the AFM Control software has a collapsible navigation area to the left. Additionally, the software utilizes a tab-system containing three (or more) tab-screens that can be viewed by pressing on the tab at the top of the screen. The first tab is for the Pre-Scan Window (Section 3.2.1), Topo Scan Window (Section 3.2.2), and Force/Distance Window (Section 3.2.4). The first two tabs (Pre-Scan and Topo Scan), along with the Sample Navigation Area (Section 3.2.3), are all that are needed for measuring AFM images. If additional modes are installed on the system (e.g. Magnetic Force Microscopy or Lithography), there may be more tabs available to select.

### 3.2.1 Pre-Scan Window

The Pre-Scan Window contains all of the functions that require adjustments before an AFM image is measured. Normally, this is used prior to any other parts of the software. This section is used to set-up the instrument's parameters, such as Photoluminescent Diode (PLD) alignment, Range Check, and Tune Frequency, in addition to approaching the probe tip to the sample surface. In this window, the various controls are organized in an order which follows from top-to-bottom and left-to-right.




#### 3.2.1.1 Modes

There are two image scanning modes available with the AFM: Vibrating (Tapping) and Non-Vibrating (Contact) mode. The modes are selected with the "Scan Mode" button. When Vibrating Mode is selected, the Tune Frequency window is activated. When Non-Vibrating mode is selected, the frequency sweep window appears grayed out and is deactivated.

#### 3.2.1.2 Modes

The TT-2 AFM is designed with care in order to enable flexibility in it's ability to scan at various resolutions. In order to simplify this process, there is a section of the Pre-Scan tab labeled 'Resolution,' with options ranging from High, Higher, Highest, and Custom modes. These controls allow the user to choose a trade-off between large scan sizes and ultimate resolutions. The 'high' resolution mode will limit the ultimate resolution of the instrument, but

has great resolution, it will artificially limit the area and feature heights that can be scanned. The 'custom' mode allows the user to setup their own HV gain settings for maximum flexibility. To better understand these parameters and adjustments, see Section 6.

**NOTE: The 'High' resolution mode does allow for high-resolution imaging, and can be successfully applied to image features as small as 5 nm. In addition, this mode is the simplest to use and allows for large AFM scans. Therefore, unless the user is expecting to measure features with heights below 5nm in size, it is recommended to use the 'High' resolution mode.**

**NOTE: As a convention, each control cluster in the AFM Control Software has a *settings* window for advanced parameter controls, or to change the way these controls work where available. This can be accessed by pressing the gear in the top right hand corner of the panel.**



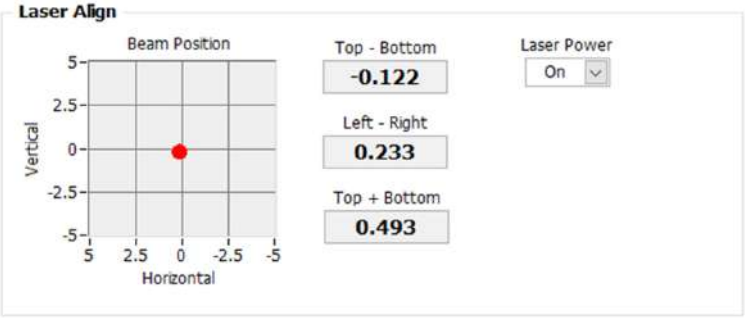
**Settings**: The settings window for the Resolution modes can be accessed by clicking the gear icon. This window allows changing various settings for each of the four available Resolution modes. In general, the user will only want to change the X and Y HV Gain and Z HV Gain values. We recommend that only the 'Custom' values are changed, as it is best to leave the 'High,' 'Higher,' and 'Highest' settings at their default values, as shown below.




### 3.2.1.3 Photoluminescent Diode (PLD) Align

The position of the Photoluminescent Diode (PLD) on the four-quadrant photodetector is presented both numerically and visually in the Photoluminescent Diode (PLD) Align Window. These two indicators are both updated in real time and are used for aligning the light lever. There is also a switch for turning the Photoluminescent Diode (PLD) on and off, useful for highly reflective samples ***Please note that the Photoluminescent Diode (PLD) should be left on when motoring towards and tip approaching to sample surface*** In addition, the 'Top+Bottom' signal is displayed, signifying the total Photoluminescent Diode (PLD) signal, or saturation, registered by the photodetector.

**NOTE: After setting up the TT-2 AFM Light Lever for the first time (see Appendix A.6), the thumb screws used for Photoluminescent Diode (PLD) alignment should not need to be turned more than a few turns.**




### 3.2.1.4 System Panel

The system panel has a drop-box for selecting the scanner to be used by the TT-2 AFM. The TT-2 AFM instrument uses a sample-scanning configuration. The types of scanner that can be selected are:

- SS 15: This is a high resolution scanner that has a range of 15 x 15 µm in X and Y and 7 µm in Z. Unlike the other sample scanning scanners, the SS 15 scanner has no position sensor in the Z Axis.

- SS 50: This is a highly flexible mid-range scanner that has a range of 50 x 50 µm in X and Y and 17 µm in Z. The SS 50 scanner has position sensors in all three axis.

- SS 100: This is a large-range scanner that has a range of 100 x 100 µm in X and Y and 17 µm in Z. The SS 100 scanner has position sensors in all three axis.

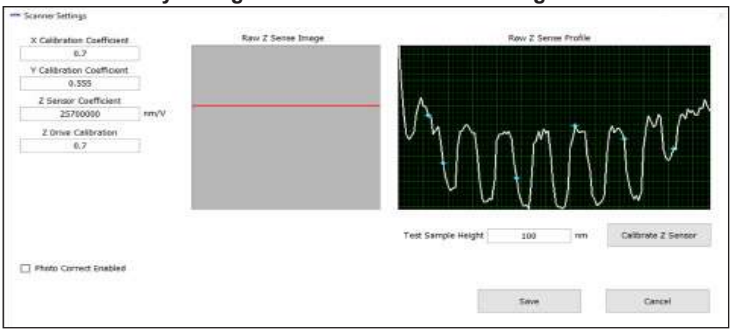
Just below the drop-down box for selecting the scanner is the Range Check button. The usable range of the piezoelectric scanner is measured when using this Range Check option. When started, the scanner is moved in a square pattern (in X and Y), which can be readily observed with the video optical microscope, and then up and down in Z. This function must be completed before beginning a scan. A Range Check MUST be performed everytime a new scanner or resolution mode is selected. Until the Range Check is performed and the "Range Check Valid" light is shown, the AFM Control Software will not allow scanning to take place.



**CAUTION: Before activating the Range Check function, make sure that the probe is moved away from the sample surface. If the probe is too close to the surface, the probe may be damaged.**

**Settings**: The settings window for the System frame is shown below. This window allows the user to change the calibration values for the selected scanner. Appendix C explains how to calibrate the TT-2 AFM in detail. There is also an Orthogonality Error correction feature, useful for non-linearity in a scanner. This adjustment allows one to linearize images.

**NOTE: Changing the calibration values will change the scale on images measured with the TT-2 AFM. Only change the values when calibrating with a reference sample.**




### 3.2.1.5 Tune Frequency

The Tune Frequency window is used for selecting the optimal conditions for producing vibrating-mode images. There is one oscilloscope window with two traces: the blue trace shows the Vibrational Amplitude of the probe's vibration while the red trace shows the phase between the drive and measured frequencies. The variable controls are:

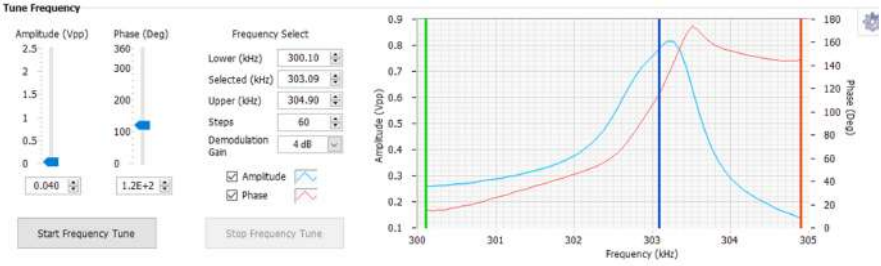
1. ***Amplitude*** and ***Phase Sliders***: Adjusting the amplitude scroll bar will alter the driving amplitude for the probe's vibration. Adjusting the phase scroll bar alters the phase degree. For optimal scanning, the amplitude should not exceed 2.5V or the resonance will become saturated.

2. ***Lower***, ***Selected***, and ***Upper Frequency***: The lower, selected, and upper frequencies correspond to the green, blue, and red vertical bars in the oscilloscope windows respectively. When trying to find the resonance peak of a new probe, it is advised to set the lower and upper frequencies to the boundaries of the frequency of the probe type, which should be found on the box or in the accompanying literature. If not, set the lower and upper bounds +/- 25 KHz from the stated frequency. The selected frequency should fall just to the left of the maximum of the resonance peak and on the steepest part of the slope of the phase curve, owing to attractive and repulsive forces at work on the cantilever's oscillation.

3. ***Steps***: This controls the number of times the amplitude and phase are sampled as the frequency is swept. For larger ranges, this number should be larger to avoid 'missing' the frequency peak. One step per kHz is a good starting number, but to see details of the peak, larger numbers can be used.

4. ***Start*** and ***Stop***: These buttons can be used to start and stop frequency sweeps.

5. ***Demodulation Gain***: This function adjusts the gain used in the detection of the oscillating signal from the photodetector. 4dB is a good starting value for a reference sample, but this should be altered in conjunction with the drive amplitude with different probes and samples to maintain a resonance peak under 2.5V. Note that while larger values may be necessary in some cases, maintaining this value below 13 dB usually leads to a lower noise and increased stability in the system.

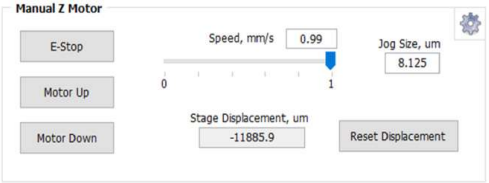



### 3.2.1.6 Manual Z Motor Control

The Z motor, which raises and lowers the light lever, is activated using the controls in this window. The speed of this motion can be controlled with the speed slider. When the Up and Down buttons are held down, the motor will move the designated Speed mm/s. When the Up and Down buttons are pressed momentarily, the motor will move the designated Jog Size um. The Stage Displacement display shows how far the motor has moved. Using the Reset Displacement button when the probe is on or near the surface will reset this Stage Displacement display, allowing one to keep better track of their distance above the sample surface.

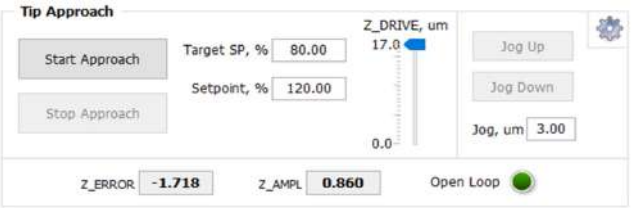


**CAUTION**: **Always visually monitor the position of the Z motor. If not monitored, the Z motor can drive the cantilever into the sample stage and can cause damage to both the probe holder and scanner.**




### 3.2.1.7 Automated Tip Approach

The Automated Tip Approach routine is a safe method to approach the surface and put the system into feedback without risk of damage to the probe. A woodpecker motion is used, which results in noticeable clicking sounds from the microscope stage. This sound is from two sources: the stepper motor being energized and the Z piezoelectric ceramic being retracted. When the system is engaged in feedback, the Open Loop indicator will light up and change to “In Feedback.”

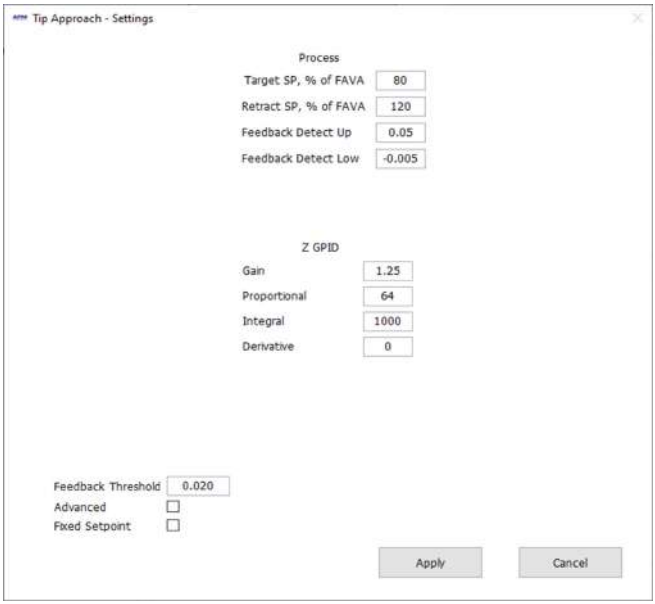



The controls to the right of the Start and Stop Approach buttons allow the user to alter the Target Setpoint and current Setpoint. The control to the right of this is a Z_Drive display, which shows the live extension of the Z Piezo. The movement of the blue arrow up and down gives a visual display of the progress of the tip approach routine.

On the far right, there are controls that are activated when the system is in feedback. These controls allow the user to safely manipulate the extension of the Z Piezo. The Jog, um box allows the use to choose a distance to jog the blue arrow's position up or down between 0.33 and 3.00 microns by using the Jog Up and Jog Down buttons.

At the bottom of this panel are displays which allow the user to see the current state of the approach routine (Open Loop or In Feedback) as well as a live read-out of the Z_Error and Z_Ampl signals. In Vibrating Moode, the Z_Ampl value signifies the current vibrating amplitude, while in Non-Vibrating mode, this displays the Top-Bottom photodetector signal.

**Settings:** The default settings window for the Tip Approach panel is shown below.

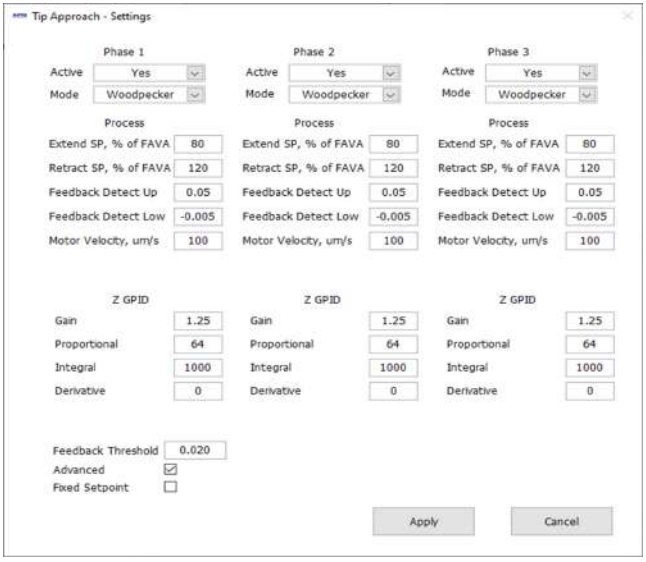


This window has various controls which can alter the way that the automated tip approach process operates. In general, the user will not need to alter these values from their default values, which are shown above.

Note that the settings in this window will have a slightly different appearance in vibrating and non-vibrating mode. The user may want to alter these with very soft and or dirty samples, or when using new probes and struggling to get into feedback. This can also be used to speed up the approach by adjusting the Z GPID.

The first two values are those used to move the probe closer to the surface using the Z Piezo, and away again, in the woodpecker method. NOTE: FAVA stands fro Free-Air Vibrational Amplitude. The next two values relate to the values used to detect when the probe is in feedback. The GPID values are those used to move the probe, in which larger values will speed up the tip approach. The feedback tTT-2eshold values reflects how large the error signal can be for the system to consider the feedback loop to be engaged correctly. Noisy imaging may require a larger value to maintain feedback.

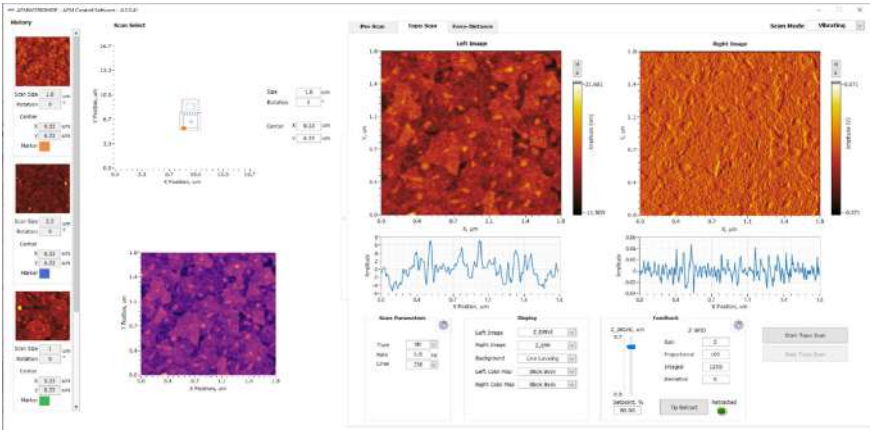
The Fixed Setpoint option allows a feedback system where the system will always approach the Target Setpoint, regardless of how many times the approach button is pushed, and should be used with caution. The Advanced check box enabled a more complex approach mechanism for very fine control of the tip-sample forces during approach. The default values for this advanced routine can be seen in the window below.




The advanced setting for probe approach is shown above. The three phases allow the user to approach quickly until relatively close to the surface, then proceed slowly as feedback is approached, allowing fine control of the tip approach forces. Reducing probe-sample forces during approach can enable the use of highly sharpened probes without damage. See Chapter 6 for an in-depth explantation of high-resolution imaging.

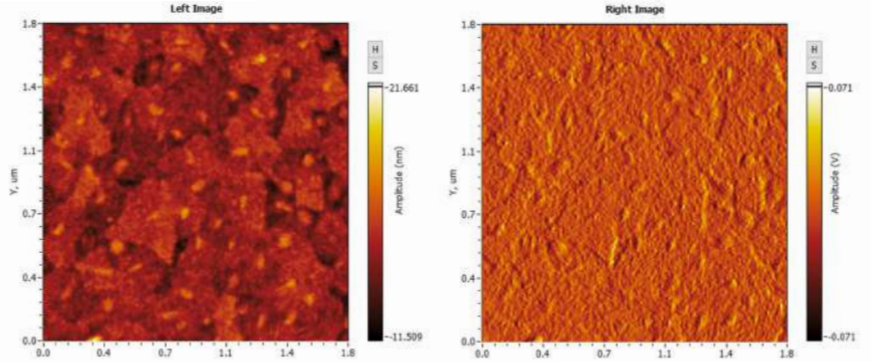
### 3.2.2 Topo Scan Window

After the Pre-Scan functions are all setup and tip approach has completed, we will move on to the Topo Scan window. This window is used for measuring AFM images.



### 3.2.2.1 Image Windows

Two image channels are displayed simultaneously while scanning. The type of image data and their appearances are selected and controlled in the Display Panel. As the images are displayed, there is a constant normalization of the data so that the images display at the full Z Axis scale.




The grey H and S buttons on the right side of the Image Windows allow users to open the scans in a larger window (by pushing S) and adjust the histogram in real time (by pushing H).



The histogram (H) can be adjusted by moving the red and green vertical bars. The scan (s) button opens a larger version of the image in a new window.

### 3.2.2.2 Oscilloscope Windows

A z-height vs x-position plot of the probe as it scans across the surface is plotted in two graphs below the image scans. The data plotted is selected in the Display Window. These line profiles can be useful to estimate the height of features being scanned, as well as aiding in feedback optimization.

### 3.2.2.3 Z Feedback

All of the parameters for maintaining feedback are provided here. These parameters are:

- Gain: Scale factor for the feedback control signal

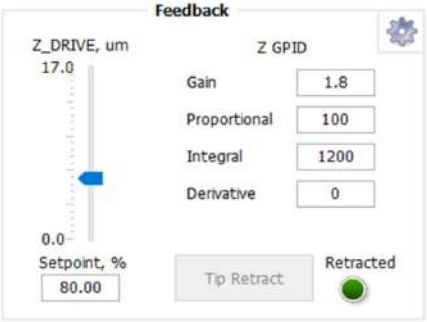
- Proportional: Scale factor for the proportional term in the PID feedback controller.

- Integral: Scale factor for the integral term in the PID feedback controller.

- Derivative: Scale factor for the derivative term in the PID feedback controller.

- Setpoint: Parameter that controls the magnitude of interaction between the probe and surface. In the case of non-vibrating mode, larger values increase the force of interaction, while in vibrating mode, lower values indicate stronger interactions.

See Appendix G for advice on setting up GPID parameters.




**CAUTION**: **If the Z Feedback parameters are too high, the Z ceramic can begin to oscillate and potentially cause damage to the ceramic.**

### 3.2.2.4 Display Window

There are several options for changing the appearance of the images dipslayed in the Scan Window. These are:

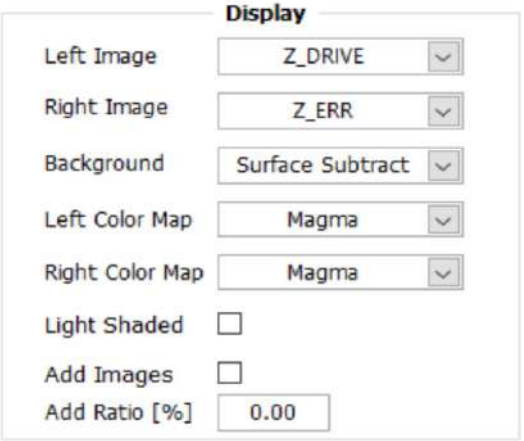
- Left & Right Image: The Source of the image displayed in the left or right window.

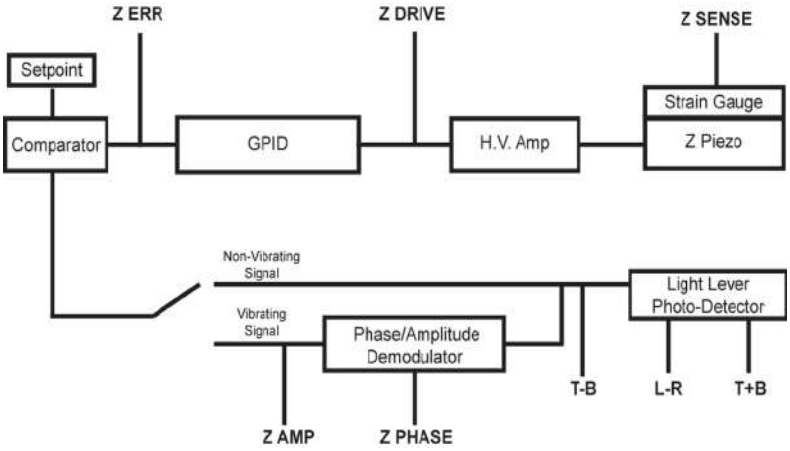
- Background: Leveling techniques used to improve the in-software display of the AFM image. This leveling change will NOT be saved in the saved AFM data, only the No Background option will be saved as raw data.

- Left & Right Color Map: This menu allows the user to select the color scheme used for the false color maps in the left and right channels.

- Light Shaded: A visual technique meant to provide a more photo-realistic view of the surface topography. This technique can add additional contrast to an AFM scan. Please note that similar to Background, this data is not saved and is a visual change only.

- Add Images / Add Ratio %: A visual technique that allows the user to 'add' a portion of the right image channel to the left image. This can be useful to add more contrast or improve areas of higher error, however this works with all available channels. Please note that similar to Background, this data is not saved and is a visual change only.



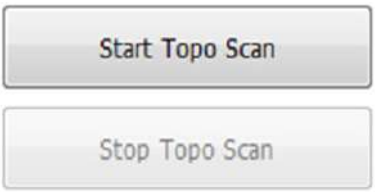



### 3.2.2.5 Tip Retract

After scanning a sample, this function is used to move the tip away from the surface safely. The Tip Retract function should always be used to assure that the probe does not get damaged after scanning. This should be performed once scanning is completed and the user wants to move to a different section of the sample or prior to motoring up for tip or sample exchange. Note that after a successful Tip Retract, the Retracted light will be illuminated in light green and the probe will have moved roughly 50 µm from the surface.

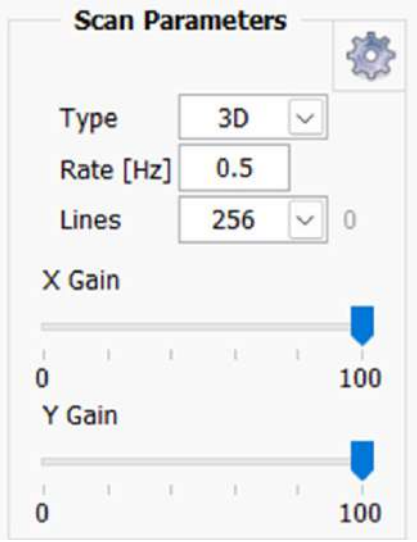
### 3.2.2.6 Scanning

The Start Topo Scan and Stop Topo Scan buttons allow the user to start and stop scans. At the end of each scan, or if the Stop Topo Scan button is activated, the user will be prompted to save the images recorded.

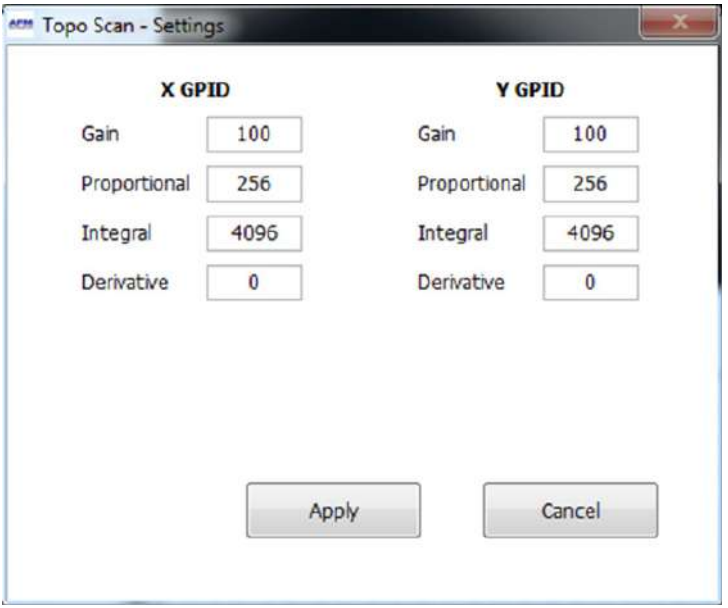



### 3.2.2.7 Scan Parameters

This panel includes the 2D/3D control which allows the user to toggle between line scanning, and full image scanning. It also allows control of scanning rate (lines per second), and the number of lines per image (the digital resolution of the image).




Settings: The parameters on the Scan parameters settings page, control the feedback between the position sensors of the TT-2 AFM and the X-Y piezos. In effect, the y control the linearity of the resulting scan. It is recommended to only change the Gain parameter between 0 and 100 %. 0 % will give a totally non-linearized image, but enables the lowest levels of noise. 100 % gives a fully linearized scan, with moderate noise, which will be visible in small scans.



### 3.2.3 Navigation Area

The navigation area is a pop-out panel on the left of the main screen of the AFM Control software. It has two main areas, a narrow vertical History strip on the left, and the Scan Select area on the right. The Scan Select panel houses the main controls used to select the area of the sample to scan. The top square area shows the total range of the currently selected scanner. The controls on the right allow the user to select the scan range and the scan angle from 0 to 360°. Either controls can be change, by typing the desired value into the box, or by placing the cursor in the box and using the arrow up and down buttons. Below this are the "Center" controls, these control the location of the center of the area to scan. Again, they can be changed by typing, or using the cursor (the scan range must be larger than the maximum range of the scanner. In addition, this can be controlled by clicking in the scanner range, to drag the scan area center to the desired location. The blue dotted box in the Scan select area shows the currently selected scan area, with a cross in the scan center, and the orange dot shows the starting location of the probe.

The History strip on the left of the navigation area holds the locations (center X and Y locations, scan size, and scan angle) of the previous scans made. The images saved (left image channel) are displayed in the history strip as small thumbnail images. Note that unsaved images do not appear here. Each of the thumbnails is associated with a colored marker, which shows the location of the previous scan in the scan select area. Clicking on any of the thumbnail images will load the scan size, location, scan angle and resolution back into the scan controls. This is particularly useful for zooming in and out of features within the scan range.

### 3.2.4 Force/Distance Curves

Force/distance curves are created by measuring the deflection of the cantilever as the sample is moved towards and then away from the probe at the end of the cantilever. The shape of a F/D curve depends primarily on the cantilever stiffness and material, the nature of the sample, and the thickness of a surface contamination layer or other environment. For information about how to analyze F/D curves, see Appendix I.



The AFM must be in feedback in non-vibrating mode before a F/D curve is measured. There are three important parameters that must be set before the F/D curve is measured. They are:

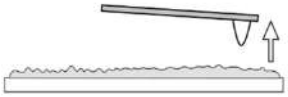
1. Down: The distance in nanometers that the sample will be moved away from the probe when the F/D curve measurement is initiated.

2. Up: The distance in nanometers that the sample will be moved up from the feedback position during the F/D curve. During this part of the curve, the probe is touching the sample surface.

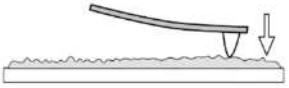
3. Rate: The rate or velocity of the sample as it is moved towards and away from the probe.



Feedback position.
Probe starts here, and returns to this position after F-D curve



Move up.
The probe moves the specified distance up before F-D curve

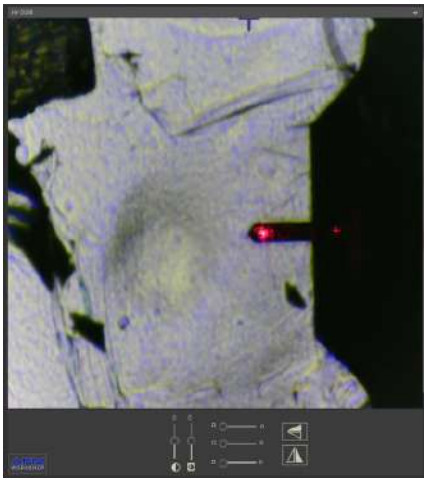


Move Down.
The probe will move the specified distance down before reversing

**NOTE: The advanced Force-distance software from AFMWorkshop enables considerably more detailed control over force-distance curves. The use of the advanced software is described in a separate User Guide.**

### 3.3 Workshop Video Software

The Workshop Video software allows real time visualization of the probe and surface.. It is recommended to optimize the parameters for each type of sample. Before operating the video camera, it must be properly aligned; see Appendix A.



The quality of the image viewed with the top view video microscope depends directly on the reflectivity of the sample being viewed. Therefore, it is not possible to give instructions for optimizing the video microscope images. It is recommended that operators experiment to find the best parameters.

## 4. Measuring Images with the TT-2 AFM

Once the TT-2 AFM is assembled, users can expect to take a few hours to learn how to measure images. It is very important to learn how to:

1. Operate the video microscope.

2. Prepare samples.

3. Exchange probes.

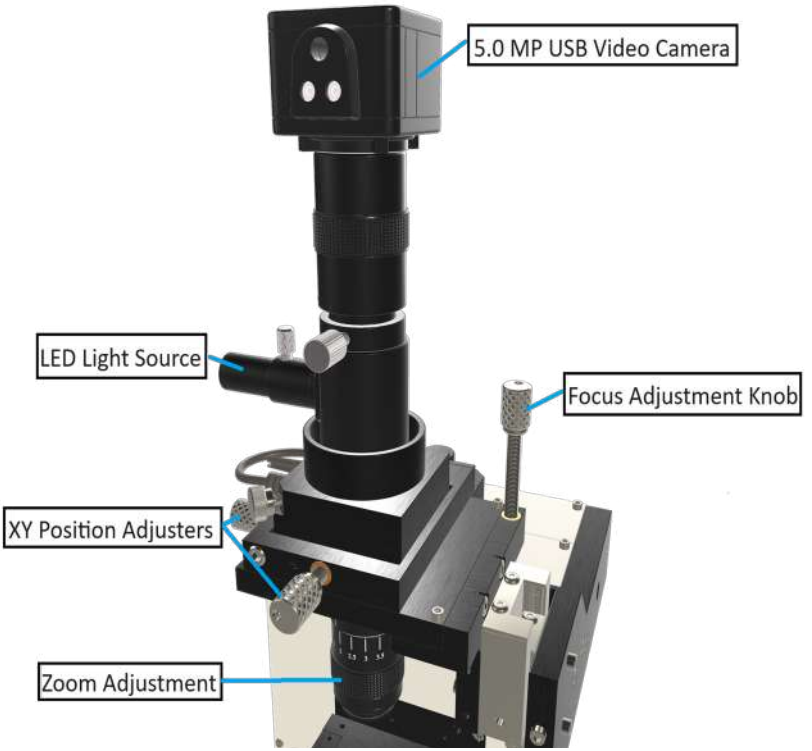
4. Align the light lever force sensor.

5. Perform video-microscope-assisted tip approach.

6. Once these techniques have been mastered, measuring images is relatively simple.

### 4.1 Operating the Video Microscope

Operating the video microscope is essential for efficient operation of the TT-2-AFM. It allows for proper viewing of the sample, assists in manual tip approach, and is critical for Photoluminescent Diode (PLD) and photo-detector alignment. It is advised to practice operating the video microscope before attempting to replace probes or measure images. The components of the video microscope are:




1. ***XY Position Translator:*** These consist of XY position translators used for centering the video microscope window over the probe location.

2. ***Zoom Adjust:*** The zoom lens allows for viewing of a surface at high or low magnification. The light intensity and focus may need to be adjusted when making large changes with the zoom lens.

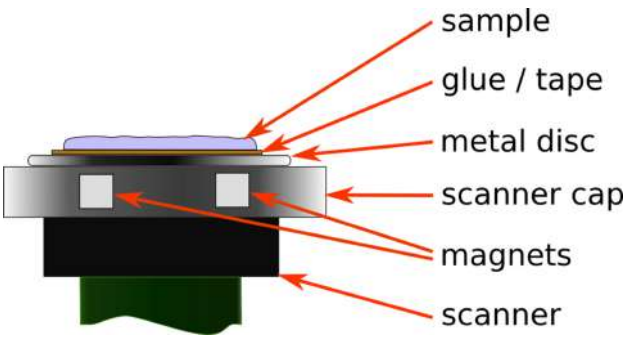
**3.** ***Focus Adjustment Knob:*** This allows for the Z adjustment of the camera assembly, allowing you to adjust the focal point of the camera to focus on the sample or probe.

4. ***Light Intensity Adjust:*** It is often critical to adjust the intensity of the light on a sample to get the best image. This is controlled via a separate light-adjustment control box (not shown).

**NOTE: Since the video microscope works using reflected light, the reflective quality of the sample can affect the quality of the image obtained.**

### 4.2 Preparing Samples

Metal disks serve as sample holders for the TT-2-AFM. Attach the sample to a metal disk with double-sided sticky tape or glue. Once a sample is attached to the disk, it is held onto the microscope's sample holder magnetically. It is critical that the sample be tightly bound to the sample disk in order to avoid vibrations, which degrade the image resolution.




#### 4.2.1 Exchanging Samples

Follow the steps below to successfully exchange samples with the TT-2 AFM.

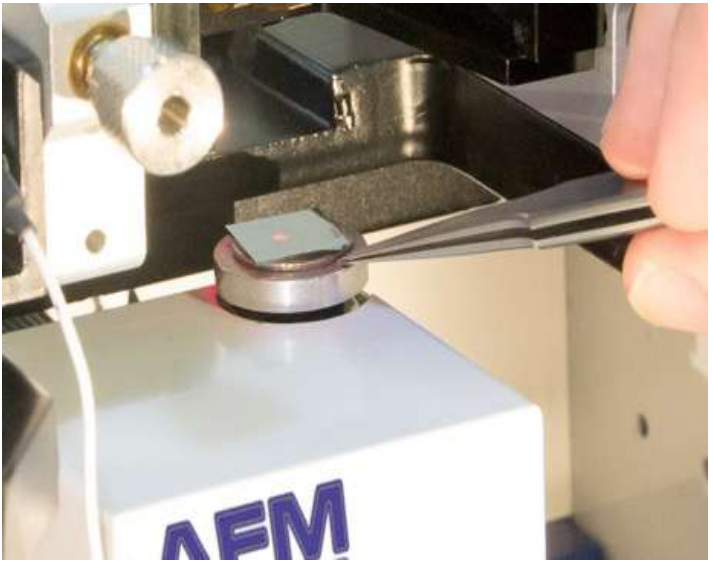
1. Raise the Z motor until the probe is at least ¼″ from the sample's surface.

2. Remove the probe holder (see section 4.3).

3. Place the sample disk on the top of the sample holder. Adjust with small, sharp tweezers until the sample is visible in the video microscope.



**CAUTION: Do not place excessive force on the sample holder, as it could break away from the piezoelectric ceramic.**



4. Align the sample using the video optics (see section 4.1).

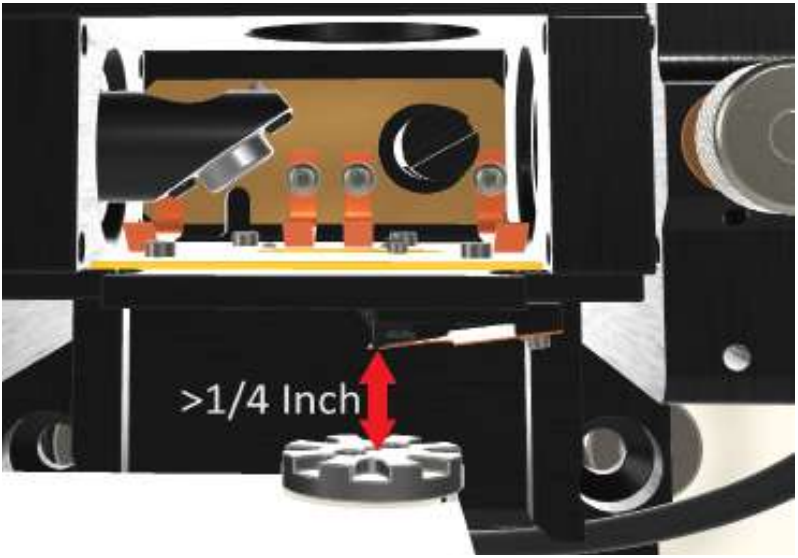
5. Replace the probe holder with the probe tip facing the sample.

### 4.3 Exchanging Probes

Metal disks serve as sample holders for the TT-2 AFM. Attach the sample to a metal disk with double-sided sticky tape or glue. Once a sample is attached to the disk, it is held onto the microscope's sample holder magnetically. It is critical that the sample be tightly bound to the sample disk in order to avoid vibrations, which degrade the image resolution.

When a tip becomes contaminated, damaged or broken, these steps can be followed to exchange probes quickly and efficiently.

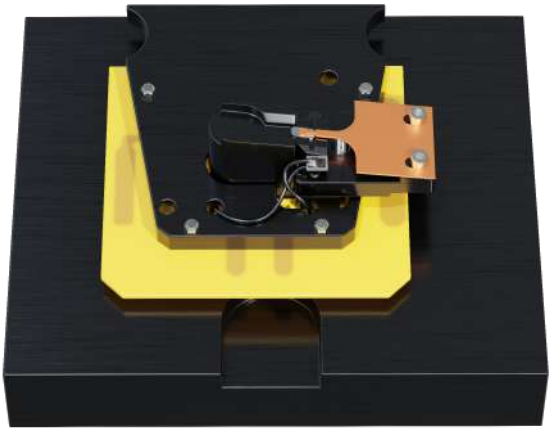
1. Raise the Z motor until the probe is at least ¼" from the sample's surface.



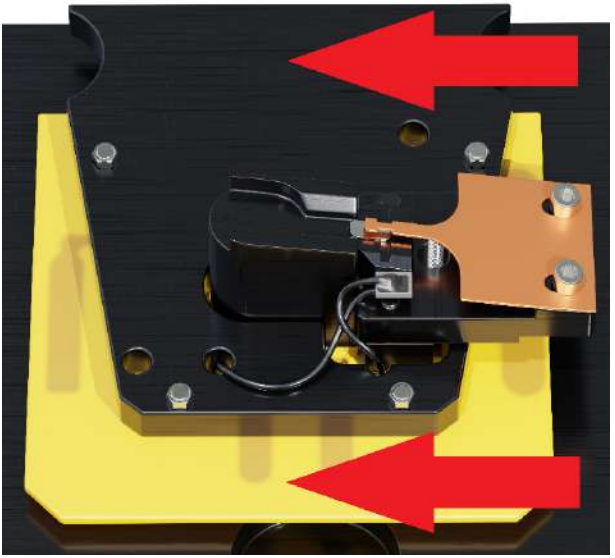
**NOTE: Make sure the new probe is close at hand and facing in the same orientation as the probe exchange tool.**

2. Remove the probe holder from the light lever.

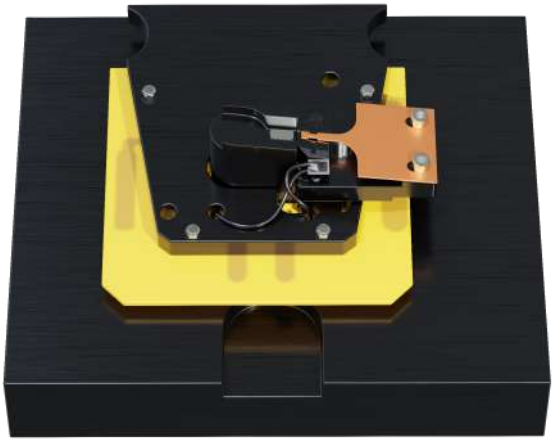
3. Place the probe holder on the probe exchange tool with the spring facing up.



4. Push down on the probe holder on the sections outlined below, such that the spring releases the probe.



5. Tap the probe exchange tool lightly on a flat surface at a 30° angle until the probe slides down through the narrow channel of the exchange tool. Gently dispose of the old probe with tweezers.



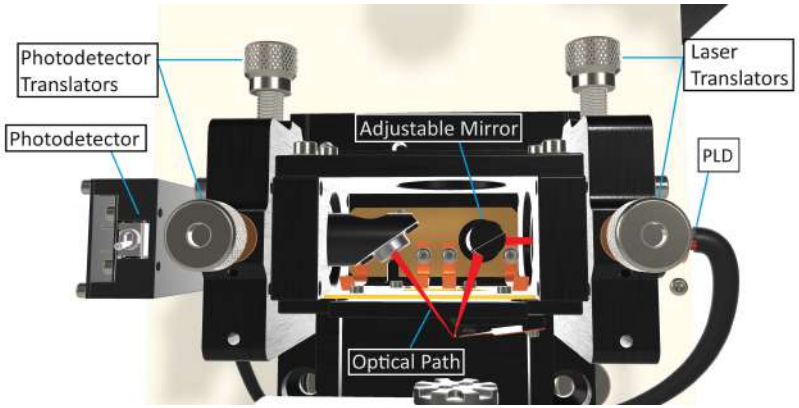
**CAUTION:The exchange tool will not prevent expulsion of the probe if the tapping angle is too extreme.**

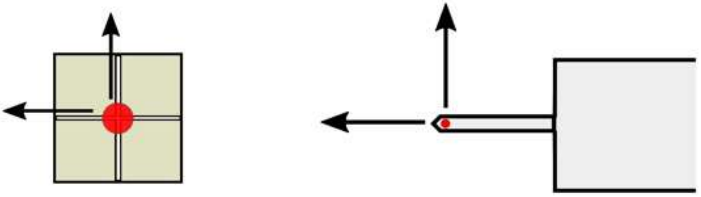
6. Using the tips of very sharp tweezers, pick up the new probe and, while pressing down on the probe holder, gently slide the probe into place. Take care not to touch the end of the chip with the cantilever.

**NOTE: If the probe holder does not sit completely level with the probe exchange mechanism, it is possible to adjust to the screws on the bottom of the exchange tool to achieve the desired height.**

### 4.4 Aligning the Light Lever Force Sensor

Aligning the light lever force sensor is greatly facilitated by understanding the optical path of the light in the sensor. The following figure shows the light path and the effect of turning the adjustment knobs on the light lever.



Motion of the laser spot when the photodetector adjust knobs are turned clockwise

Motion of the laser spot when the laser adjust knobs are turned clockwise

### 4.4.1 Tips for Aligning the Light Lever Force Sensor

1. Bringing the cantilever within a millimeter of the sample's surface can help the user see the Photoluminescent Diode (PLD) on the sample.

2. After replacing a cantilever, the Photoluminescent Diode (PLD) and photo-detector adjustments are usually less than a single turn.

3. The adjustment screws may have backlash; sometimes the motion is not perfectly linear.

4. If the cantilever does not align within a few minutes, try another cantilever.

5. The detector has two modes: high and low gain. High gain allows more Photoluminescent Diode (PLD) light through the detector than low gain. High gain is commonly used for most scanning purposes because it has a better signal to noise ratio.

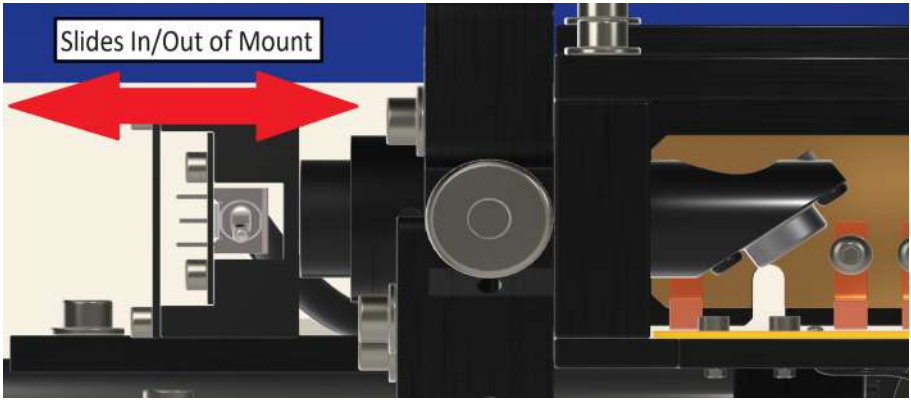
6. When in vibrating mode, if the resonance peak appears inverted at the top, reduce the amplitude slider. If further adjustments are necessary, switch the detector gain from high to low.

7. If the Beam Position is not aligned within a few turns of the adjustment knobs, the photo-detector can be manually adjusted by manually gripping the photo detector, and sliding it a small distance in our out of the light lever head. Repositioning the detector so that the Top + Bottom signal is maximized while the beam position is mostly centered. Normally some small adjustment with the photodetector screws will be required after this operation.

**NOTE: In normal use, there is not requirement to slide the photodetector in or out of the light lever head. This may become necessary in two situations:**

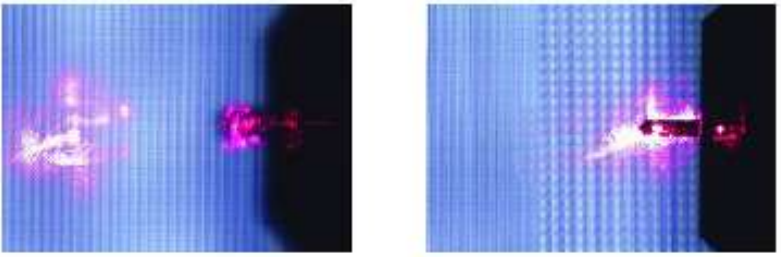
**1. When changing from a very long to very short type of probe, or vice-versa.**

**2. When changing from air-scanning to scanning in liquid or vice-versa.**




### 4.5 Optically-Assisted Tip Approach

Tip approach with the TT-2 AFM is achieved using a woodpecker algorithm activated in the Pre-Scan window. To save time during tip approach, move the probe as close to the surface as possible before activating the Automatic Tip Approach function. The video microscope can prove very helpful for manual probe approach. Below are two images of a probe as it moves toward a standard reference sample surface.



Probe far from surface (left), and probe close to surface (right).

### 4.5.1 Performing Optically-Assisted Tip Approach

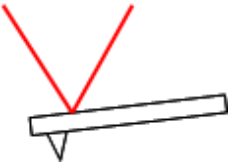
1. Focus the video microscope on the sample surface.

2. Carefully lower the Z motor with the Manual Z Motor Control, until the cantilever is visible in the video microscope image. Make sure the Jog Size has been set to a low step size (such as 5-10 µm) and the Z motor speed is set to 0.01 mm/sec in order to avoid crashing the probe into the sample.

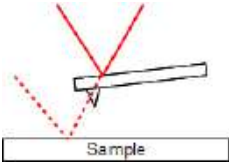
3. Slowly jog the probe down until it is only slightly out of focus.



**CAUTION: If the tip touches the surface before the feedback is activated, the tip may be damaged or even broken.**



Tip Far Away From Suface




Tip Near Surface

4. The system is now ready for Automated Tip Approach.

### 4.6 Contact Mode Imaging

These instructions assume the AFM Control Software is properly loaded and working, and that the material presented in section 4 has been reviewed.

#### 4.6.1 Pre-Scan Tab (Refer to Graphic on Page 36)

1. Follow the instructions in section 4.2. and place a sample in the instrument.

2. Follow the instructions in section 4.3. and place a contact mode probe in the instrument. See Appendix E for probe recommendations.

3. Select Non-Vibrating in the Scan Mode window.

4. Focus the video optical microscope on the AFM probe.

5. Follow the instructions in section 4.4. to align the Photoluminescent Diode (PLD) on the probe. Make sure the Top – Bottom signal is between 0.5 and 0.9 for contact-mode imaging.

6. Focus the video microscope on the sample surface.

7. If this is the first scan since opening the AFM Control Software, click on Range Check in the System window.

**NOTE: The probe must NOT be in contact with the sample surface during Range Check. The piezoelectric ceramics will move to measure the usable range, which may cause damage to the tip.**

8. Use the Manual Z Motor Control to move the light lever towards the sample (see section 4.5.).

9. Initiate Automated Tip Approach. Wait for the green light to indicate that feedback was achieved.

10. If the Z Drive indicator is above the center of its range, use the Jog Down button to move the Z Drive indicator towards the middle. You may need to change the Jog size, depending on the scanner in use (a good size is 3 µm for a 17 µm scanner, or 1 µm for a 7 µm range scanner). A few jogs may be required to reach the center of the range.

11. If the Z Drive indicator is below the center of its range, use the Jog Up button to move motor up by small increments until the indicator is at the center of the Z range.

12. Once true feedback has been achieved, open the Topo Scan tab at the top of the AFM Workshop window.



### 4.6.2 Topo Scan Tab (Refer to Graphic on Page 38)

1. Select the parameters controlling the area to be scanned; these include the scan size, the center position in x and y, and the scan rotation. Samples may be scanned between 0° or 360° by adjusting the Rotation parameter.

**NOTE: A scan rate of 0.5 Hz and 256 scan lines will achieve a basic low resolution scan.**

2. Select the Scan Rate (scanning speed, in lines per second) and the number of lines (digital resolution of the image).

3. Select the desired images to be displayed in the Display window. The Display window also allows for control of leveling. Some display options for contact (non-vibrating) mode scanning include:

- Z-Drive, the change in voltage through the z-piezoelectric ceramic, shows the sample's topography. It is common to set Z-Drive to the Left Image.
- Z-Error, the error signal, is used to help optimize the Z Feedback parameters. When initially setting up a scan, it is common to use Z-ERR as the Right Image.
- T-B signal is similar to Z-Error, and is sometimes used as the Right Image
- L-R, lateral force, detects cantilever torsion as the tip applies constant force along the surface. This mode is unique to contact mode scanning, and can only be properly used when scanning at 90°.

4. Begin a scan by pressing the Start button in the Scan window.

5. While scanning, optimize the GPID and setpoint parameters (see Appendix G).

**NOTE: When the GPID parameters are optimized, the Z piezoelectric ceramic will more closely follow the probe tip as it scans the surface. When tip movement is minimized the Z error signal is decreased. Increasing the setpoint by small increments (50-100 mV) can sharpen the image by pushing the tip harder on the surface.**

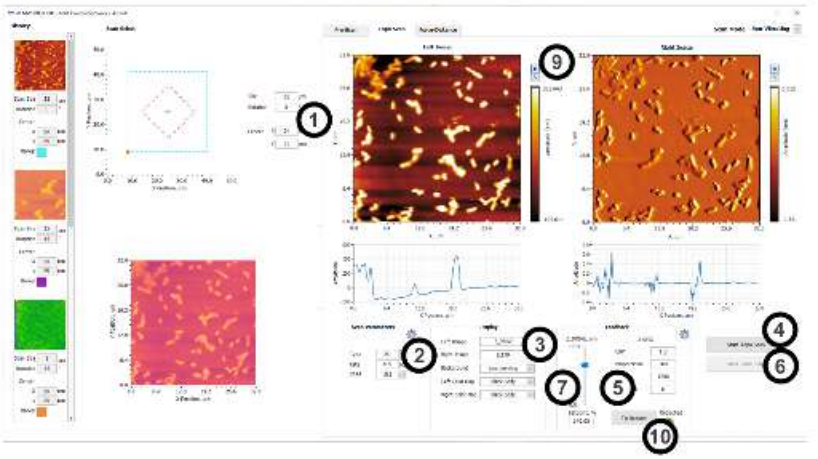
6. Once the parameters are optimized, stop the scan by pressing the Stop button. Save the image if desired.

7. Before beginning a new scan, check to see if Z Drive is still near the center of its range. If not, see sections 4.6.10 - 11.

8. Begin a new scan. When the scan is completed, the user will be prompted to save the image.

9. If desired, the user can open a larger window of both the Left and Right Images by selecting the S button to the right of the scan window. The H button allows the user to alter the histogram of the scans, which can help to visualize small features.

10. When finished scanning, press Tip Retract. This function gently releases the probe from the sample surface.




### 4.7 Vibrating Mode Imaging

These instructions assume the AFM-View software is properly loaded and working, and that the material presented in section 4 has been reviewed.

#### 4.7.1 Pre-Scan Tab (Refer to Graphic on Page 40)

1. Follow the instructions in section 4.2 and place a sample in the instrument.

2. Follow the instructions in section 4.3 and place a vibrating mode probe in the instrument. See Appendix E for probe recommendations.

3. Select Vibrating in the Scan Mode window.

4. Focus the video optical microscope on the AFM probe.

5. Follow the instructions in section 4.4 to align the Photoluminescent Diode (PLD) on the probe. Make sure the Top – Bottom signal is close to 0.5 for vibrating mode imaging.

**NOTE: The probe must NOT be in contact with the sample surface during Range Check. The piezoelectric ceramics will move to measure the usable range, which may cause damage to the tip.**

6. If this is the first scan since opening the AFM-View software, initiate the Range Check function.

7. Select the frequency under the Tune Frequency window.

a) Set the Lower limit to the lowest frequency on the probe box.

b) Set the Higher limit to the highest frequency on the probe box.

c) Set the Selected value to any frequency between the two limits. If the selected frequency exceeds either limit, the software will automatically change the values.

d) Set the number of Steps (100) and the Demodulation Gain (4 dB).

e) Click Start Frequency Tune to sweep of the selected range.

f) Move the green and red cursors toward the resonance peak to decrease the range, and place the blue cursor corresponding with the selected frequency 80% up the left side of the resonance peak.

g) Repeat steps (d) and (e) until a resonance peak no greater than 2.5V is observed. Optimal imaging can depend on the value used, with 1.0V often used as a starting point. If the amplitude of the resonance peak exceeds 2.5V, adjust the driving amplitude of vibration by altering the Amplitude scroll bar, or by directly typing a value into the box below.

h) For phase mode imaging, the resonance peak must correspond with the slope of the phase curve. To alter the phase degree, adjust the Phase (deg) scroll bar accordingly. If phase imaging will not be performed, unchecking the Phase option will remove the phase plot, making visualization of the amplitude plot easier.

8. Focus the video microscope on the sample surface.

9. Use the Manual Z Motor Control to move the light lever towards the sample (see section 4.5.).

10. Initiate Automated Tip Approach. Wait for the green light to indicate that feedback was achieved.

11. If the Z Drive indicator is above the center of its range, use the Jog Down button to move the Z Drive indicator towards the middle. You may need to change the Jog size, depending on the scanner in use (a good size is 3 µm for a 17 µm scanner, or 1 µm for a 7 µm range scanner). A few jogs may be required to reach the center of the range.

12. If the Z Drive indicator is below the center of its range, use the Jog Up button to move motor up by small increments until the indicator is at the center of the Z range.

13. Once true feedback has been achieved, open the Topo Scan tab at the top of the AFM Workshop window.




### 4.7.2 Topo Scan Tab (Refer to Graphic on Page 42)

1. Select scan parameters such as Scan Size, Scan Rate and the number of lines. Samples may be scanned between 0° and 360° by adjusting the Rotation parameter.

**NOTE: A scan rate of 0.5 Hz and 256 scan lines will achieve a basic low resolution scan.**

2. Select the desired images to be displayed in the Display window. The Display window also allows for control of line leveling. Some display options for vibrating mode scanning include:

- Z-Drive, the change in voltage through the z-piezoelectric ceramic, shows the sample's topography. It is common to set Z-Drive to the Left Image.

- Z-Error, the error signal, is used to help optimize the Z Feedback parameters. When initially setting up a scan, it is common to use Z-ERR as the Right Image.

- Z_AMPL, the Amplitude signal, which is similar to the Z-Error signal. It is quite common to use the Amplitude signal as the right image.

- Z-Phase, phase imaging mode, uses the differences in the probe's oscillation to form image contrast. The phase shifts when the probe comes into contact with areas of the sample surface with differing stiffness, adhesion, etc. Thus phase imaging mode is optimal for composite samples, those with varying surface densities and characteristics.

3. Begin a scan by pressing the Start Topo Scan button in the Scan window.

4. While scanning, optimize the GPID and setpoint parameters (instructions in Appendix G).

**NOTE: When the GPID parameters are optimized, the Z piezoelectric ceramic will more closely follow the probe tip as it scans the surface. When tip movement is minimized the Z error signal is decreased. Decreasing the setpoint by small increments (10-100 mV) can sharpen the image by pushing the tip harder onto the surface.**

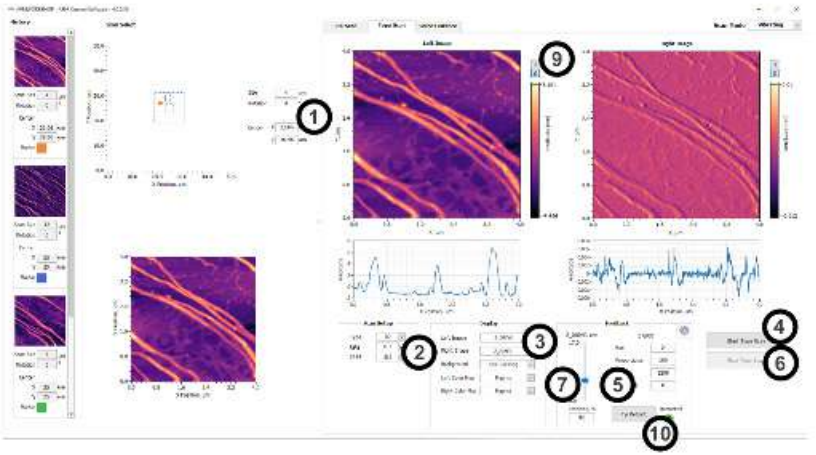
5.Once the parameters are optimized, stop the scan by pressing the Stop button.

6. Before beginning a new scan, check to see if Z Drive is still at the center of its range. If not, see sections 4.6.10 - 11.

7. Begin a new scan. When the scan is completed, the use will be prompted to save the image.

8. If desired, the user can open a larger window of both the Left and Right Images by selecting the S button to the right of the scan window. The H button allows the user to alter the histogram of the scans, making it easier to visualize small features.

9. When finished scanning, press Tip Retract. This function gently releases the probe from the sample surface.

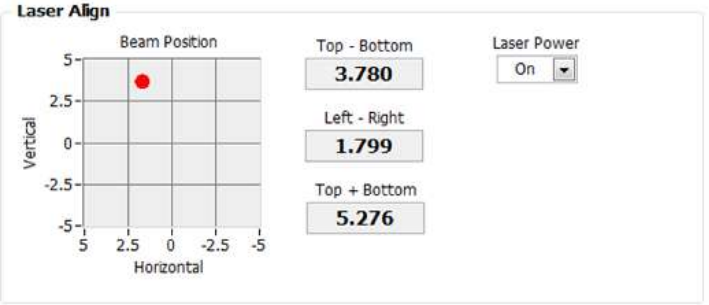



# 5. Troubleshooting – What Can Go Wrong When Measuring Images

Although the operation of the TT-2-AFM is quite simple, some problems can occur while imaging samples. The following sections detail some of the most common problems that may occur. It is important to keep these potential problems in mind when operating the TT-2-AFM.

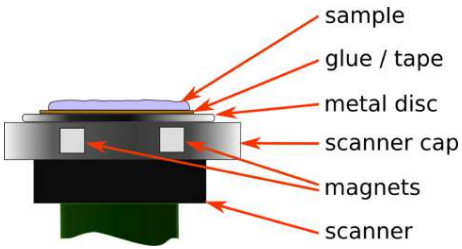
## 5.1 Saturation of the Photo-Detector

Highly-reflective cantilevers can sometimes pass too much Photoluminescent Diode (PLD) light to the photo-detector. When this occurs, the light lever force sensor will not work properly. When there is too much light, the Top + Bottom signal in the Photoluminescent Diode (PLD) Align window displays a value greater than 3. This problem can often be solved by flipping the gain switch on the photo-detector down to low gain (see section 4.4 for the location of the gain switch)



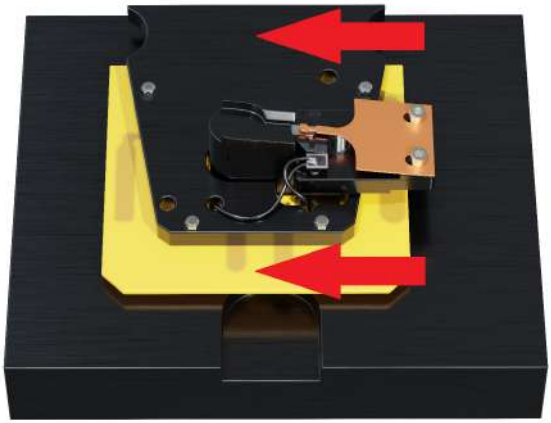
### 5.2 Sample Not Mounted Correctly

It is essential that the sample is rigidly mounted on the sample holding plate. If the sample is not mounted rigidly, it can be difficult to get into feedback, and images appear to have excessive vibrations. Equally, the sample must not move, or have loose debris that can shift during imaging.



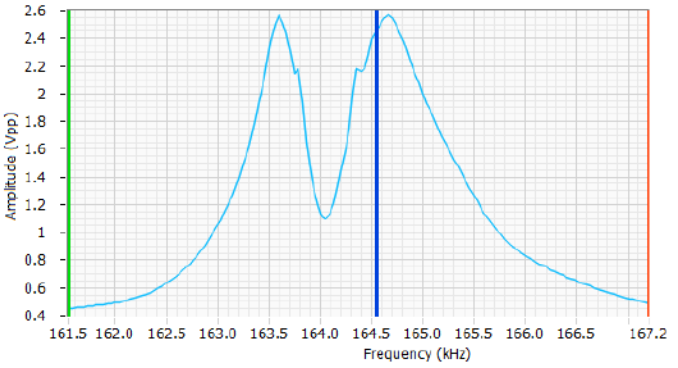

### 5.3 Probe Not Seated Correctly

If the probe is seated correctly, it will be in essentially the same place before and after the probe is removed from the AFM. If it is too difficult to align the light lever after a probe exchange, examine the probe holder closely to make sure that the probe is seated correctly. Improper probe alignment may cause resonance saturation. If the probe holder is dirty, it may also cause problems. One simple troubleshooting technique is simply pressing to release the probe from the clip, gently tapping it forward, then tapping until it falls back into place. This is referred to as reseating the probe.



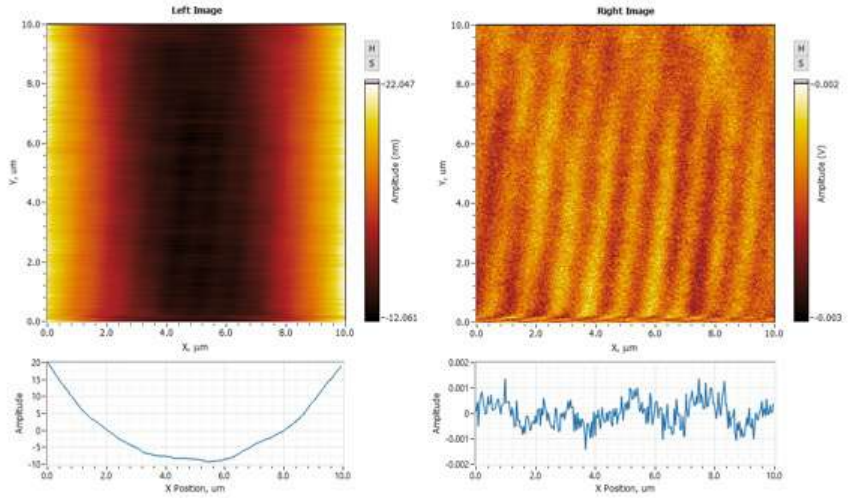
## 5.4 Resonance Saturated

When scanning the frequency of the cantilever produces a double-peaked resonance, as shown on the right, decrease the driving amplitude of the probe by lowering the Amplitude scroll bar until the resonance peak sits below 2.5V. If the resonance is still saturated, the probe may have to be re-seated.



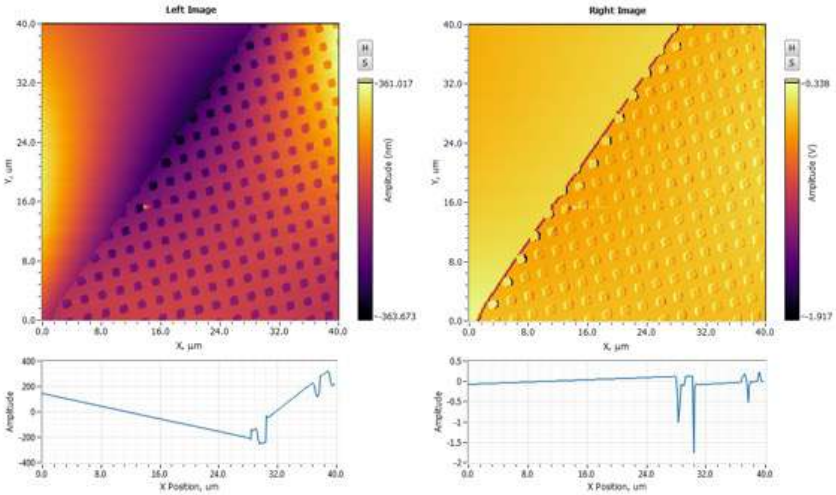

## 5.5 False Feedback

In vibrating mode, the probe can go into false feedback, resulting in wispy, featureless images. False feedback is created when the probe is caught up in the surface contamination layer. This can be corrected by reducing setpoint, activating the tip approach function a second or third time, as well as by utilizing the advanced tip approach functions to jog the tip toward the surface (see section 6).



### 5.6 Scanner at End of Z Range

When the Z ceramic expands or retracts to the end of its range while scanning, the image will suddenly lose contrast and appear flat. To solve this problem, relocate the sample so that the area being scanned lies directly over the center of the z piezo, (the screw hole of the sample holder). Next, redo tip approach so that the Z ceramic is in the center of its range, or use the Jog Up and Jog Down buttons (see section 4.6. j-k).

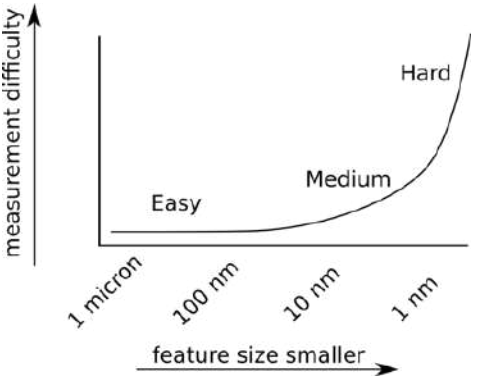



### 5.7 Defective Probe

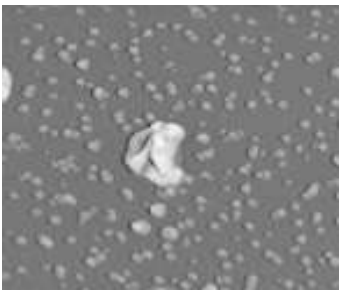
In the event that none of the above problems exist and it is not possible to get an image, the probe may be defective. Images obtained with defective probe sometimes show repeating patterns.

## 6. High Resolution Imaging

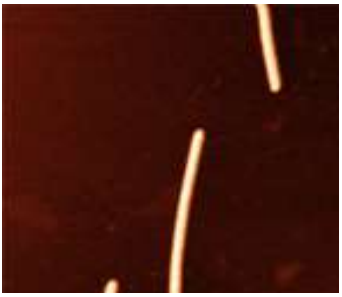
High resolution images are measureable with the TT-2 AFM. However, as with any AFM, obtaining those images requires that the AFM be optimally set up and that the operator knows the critical parameters for measuring high resolution images. The table below illustrates the difficulty in measuring images as a function of the desired dimensions.




Examples of high resolution images obtained with the TT-2-AFM include:



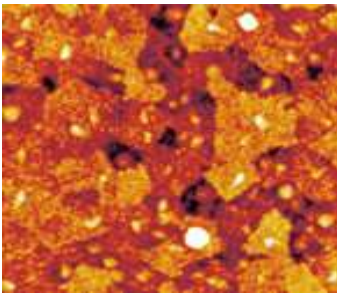
1-3 nm Test Particles



1.2x1.2 µm Tobacco Mosaic Virus



0.3 nm Si Terraces



2x2 µm Indium Tin Oxide

There are three critcal requirements for measuring high resolution images with the TT-2 AFM:

1. The instrument must not be influenced by structural and acoustic vibrations.

   For high resolution imaging, the AFM must be placed in an environment with minimal vibrations. Typically, both a vibration table as well as an acoustic enclosure is required. Often measurements must be made in the middle of the night when there are minimal structural vibrations. Measuring the noise floor of the AFM establishes the Z resolution of the AFM. Measuring images of structures with 0.3 nm in vertical dimension requires a Z noise floor below 0.1 nm. See Appendix D to learn how to measure the noise floor of the TT-2 AFM.

2. The Electronics Gain must be properly set up.

   The scanner used for the measurements must be cabable of resolving the desired features. The resolution of the scanner depends on the electronic noise in the control electronics and the mechanical gain of the piezoelectric ceramics in the scanner.

   In the TT-2 AFM the electronic noise can be reduced by reducing the Gain on the high voltage amplifiers that drive the scanner piezolectric ceramics. Alternatively, use a scanner with a small mechanical gain. The 15 micron scanner has a mechanical gain of 1:1 and the 50 micron scanner has an XY mechanical gain of 5:1.

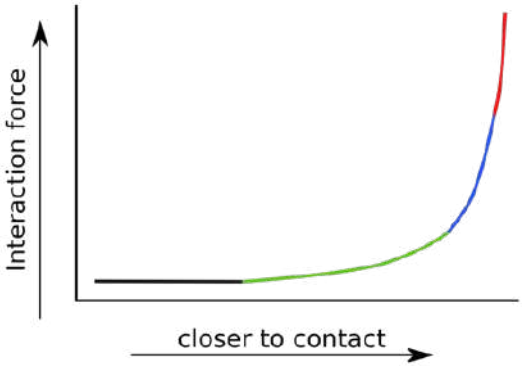
| **Theoretical Resolution For 50 and 15 µm Scanners*** | | | | |
|---|---|---|---|---|
| **Z HV Gain** | 50 µm Scanner | | 15 µm Scanner | |
| | Z Range (nm) | Z Noise (nm) | Z Range (nm) | Z Noise (nm) |
| **15** | 17,000 | 0.259 | 7,000 | 0.107 |
| **14** | 15,867 | 0.242 | 6,533 | 0.100 |
| **13** | 14,733 | 0.225 | 6,067 | 0.093 |
| **12** | 13,600 | 0.208 | 5,600 | 0.085 |
| **11** | 12,467 | 0.190 | 5,133 | 0.078 |
| **10** | 11,333 | 0.173 | 4,667 | 0.071 |
| **9** | 10,200 | 0.156 | 4,200 | 0.064 |
| **8** | 9,067 | 0.138 | 3,733 | 0.057 |
| **7** | 7,933 | 0.121 | 3,267 | 0.050 |
| **6** | 6,800 | 0.104 | 2800 | 0.043 |
| **5** | 5,667 | 0.086 | 2,333 | 0.036 |
| **4** | 4,533 | 0.069 | 1867 | 0.028 |
| **3** | 3,400 | 0.052 | 1400 | 0.021 |
| **2** | 2,267 | 0.035 | 933 | 0.014 |
| **1** | 1,133 | 0.017 | 467 | 0.007 |

AFM Workshop's 100 µm scanner has the same Z characteristics as the 50 µm scanner.

As lower Z high voltage gains are used, the maximum Z range decreases. Therefore, the step size of the Z motor must be reduced when lower Z gains are used. This will assure that the probe does not crash into the sample surface on tip approach. The software automatically adjusts the tip approach motor jog for changes in the Z HV Gain

3. Tip approach must be made such that the tip is not broken.

   Preserving a sharp tip for imaging requires making a perfect tip approach. This means that the probe must not crash into the surface and break the tip off. The best method for preserving the tip in probe approach is to bring the tip to the surface and engage in a false feedback. Then, very gently bring the tip into feedback such that the tip is interacting with the surface.



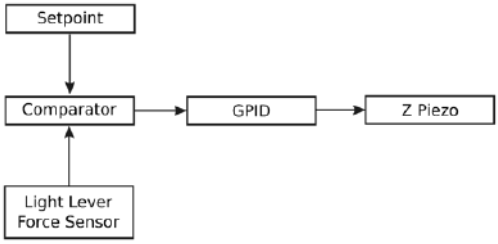

There are 4 regions of interaction forces between the tip and surface:

- **BLACK** - No interaction force
- **GREEN** - Acoustic interaction forces (also known as compression forces)
- **BLUE** - Interaction of the tip with the surface contamination
- **RED** - Direct interaction between the probe and the surface.

These interaction forces will cause the vibrating cantilever to decrease in its amplitude as it passes through each of these interaction regimes.

The method for getting into perfect feedback is to stop the tip approach in the green regime, and then slowly bring the probe into the direct interaction regime. This is achieved by setting the feedback tTT-2eshold low such that the tip approach stops when the probe is interacting with the surface contamination (In vibrating mode, Target Setpoint around 90%). Then the probe can be slowly moved towards the surface using the advanced tip approach functions. The user will know when the tip has entered the direct interaction regime when the yellow Z Drive slider no longer moves as the sample is pulled toward the tip via small setpoint changes.

The AFM control software is designed to make a careful probe approach under automated control. This is achieved with the Advanced Tip Approach window under the Pre-Scan tab in conjunction with the Tip Approach parameters in the settings window. The figure below shows how the signal from the light lever is compared to the feedback, and used to control the movement of the z piezo.



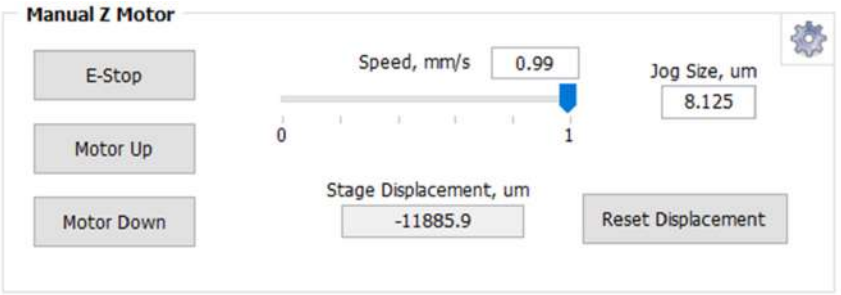

The AFM Control Software uses a “woodpecker” approach mechanism. This is a controlled stepwise movement towards the surface as explained below.

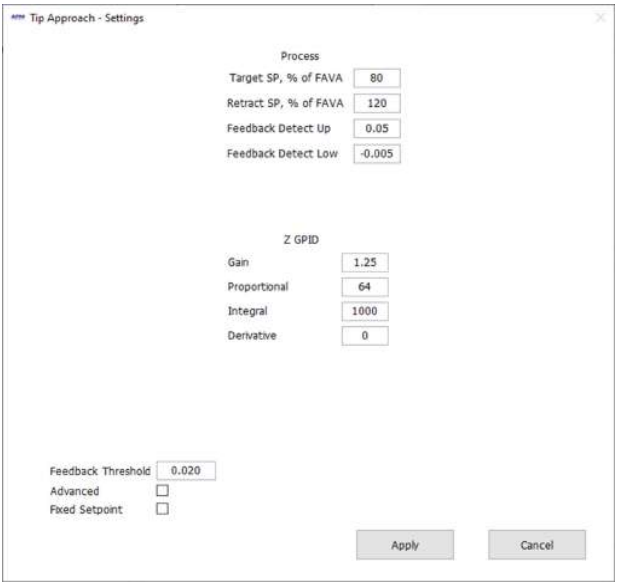
| **Tip Approach Sequence in Vibrating Mode** | | | |
|---|---|---|---|
| **Step** | **Action** | **Result** | **Tip Approach Parameter** |
| **1** | Measure vibration amplitude of the probe away from the surface | Amplitude used for initial tip approach. Set point is raised and lowered around this value which causes the sample to move up and down | Z Amplitude value in the Automated Tip Approach Window |
| **2** | Raise set point | Z piezo moves toward the probe | Extend Factor |
| **3** | Check for feedback | If in feedback, end tip approach. | Feedback Detect |
| **4** | Lower set point | Moves sample away from probe | Retract Delta |
| **5** | Activate stepper motor | Moves probe towards sample | Tip Approach Motor Jog |
| **6** | Repeat steps 2-5 | Feedback achieved with full | Z-Drive Range |

It is advantageous to move the sample up and down towards the probe at a slow rate to avoid crashing the probe into the surface. The rate at which the sample moves up and down depends on:

- The magnitude of the initial Z Amplitude
- Tip approach GPID parameters
- Target Setpoint
- Retract Setpoint

The initial value for the setpoint when feedback is established is the Z Amplitude.



# Appendix A: Setting Up the TT-2-AFM

## A.1. Selecting a Location

Structural vibrations from a building's floor and acoustic vibrations (noise) can greatly reduce the resolution of an AFM. Ideally, the AFM would be placed in a vibration-free environment, but this is typically not possible. A few tips:

- Set the AFM up in the basement or lower floor near a load-bearing wall.
- Avoid locating the AFM in the path of fans or other sources of wind turbulence.
- Avoid locations where there are elevators, a lot of foot traffic, or freeways.
- If you must place the AFM in an undesirable location, consider purchasing a vibration table and/or acoustic enclosure.

A simple test to see if a location has a problem is to fill a coffee cup with coffee and set it in the potential location. If there are ripples in the coffee's surface, there are going to be problems.

## A.2. Setting Up the Optical Microscope

A simple test to see if a location has a problem is to fill a coffee cup with coffee and set it in the potential location. If there are ripples in the coffee's surface, there are going to be problems.

The video optical microscope is essential for efficient operation of the TT-2-AFM. After setting up the microscope, spend some time learning to operate it by viewing various samples.

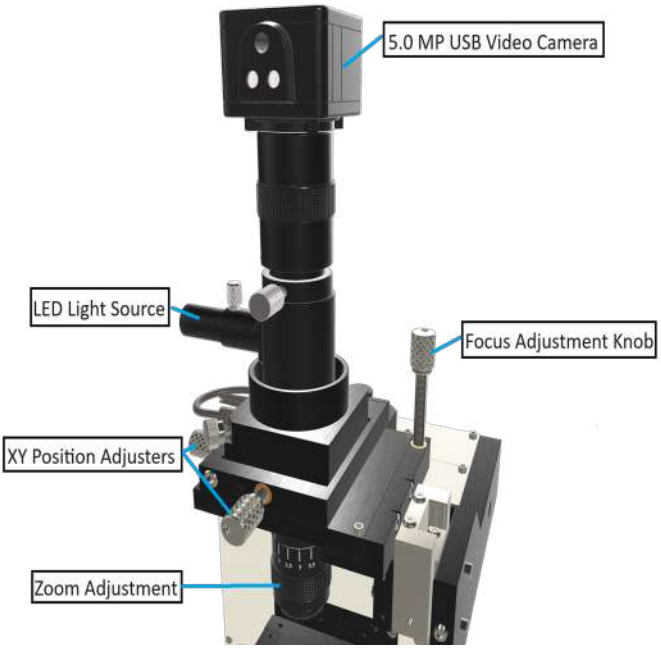
**1. Optical Microscope**

a) Install the zoom tube and CMOS camera, The USB cable receptor at the top of the CMOS camera should be facing forward

b) Attach the USB cable to the camera and computer

c) Attach the LED to the zoom tube, and connect the light power cable from the LED to the light intensity adjuster

d) Remove the black cap from the bottom of the zoom tube




**2. Test the microscope and verify its operation.**

a) Launch the Workshop Video software (see section 3.3).

b) Turn on the light and adjust the intensity to its mid-range.

c) Adjust the height of the zoom tube so that the bottom of the tube is 115 mm from the surface of the scanner.

d) Place a coin or other metal object under the objective. The light should be visible on the object.

e) On the zoom tube, turn the zoom adjustment counter clockwise all the way, thus reducing the magnification.

f) At this time, an image should be visible on the computer screen; use the focus adjust on the video microscope to sharpen the image.

g) Increase the magnification with the zoom tube adjuster and increase the light intensity as needed.

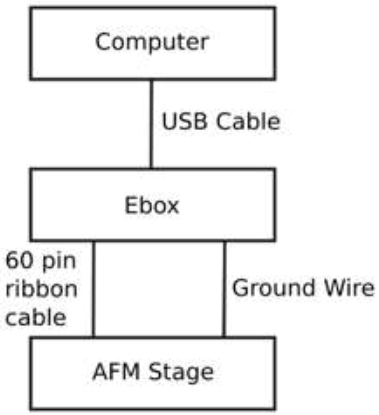


### A.3. Connecting the TT-2-AFM to the Cables

Connect the USB cable between the computer and the Ebox.

Connect the 60-pin ribbon cable between the Ebox and the TT-2-AFM stage.

Connect the grounding wire between the Ebox and the TT-2-AFM stage.



**WARNING: Failure to connect the ground wire can cause damage to the stage and/or Ebox electronic circuits.**

**WARNING: Always turn off the master power to the Ebox before plugging or unplugging any of the cables on the TT-2-AFM.**

## A.4. System Check

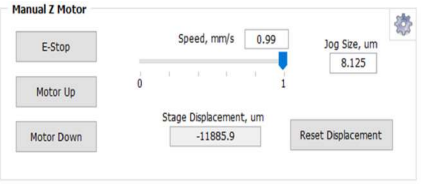
1. Turn on the computer and the Ebox.

2. Launch the AFM Control Software. The Photoluminescent Diode (PLD) should now be visible.

3. Using the Manual Z Motor Control in the Pre-Scan tab, move the Z motor a few steps.

4. If the Z motor does not respond and the Top-Bottom numbers in the Photoluminescent Diode (PLD) Align section of the Pre-Scan tab are not changing, the Ebox is not communicating with the computer. In this case, cycle power to the Ebox and restart the AFM View software.




## A.5. Aligning the Video and TT-2-AFM Microscopes

1. Raise the zoom tube on the optical microscope so that the light lever can fit under the bottom of the zoom tube.

2. Move the micrometers on the video microscope XY translators to their central position. Move the micrometers on the TT-2-AFM stage to their central position.

3.Place the TT-2-AFM stage under the zoom tube such that the hole in the light lever is directly below the zoom tube optical objective.

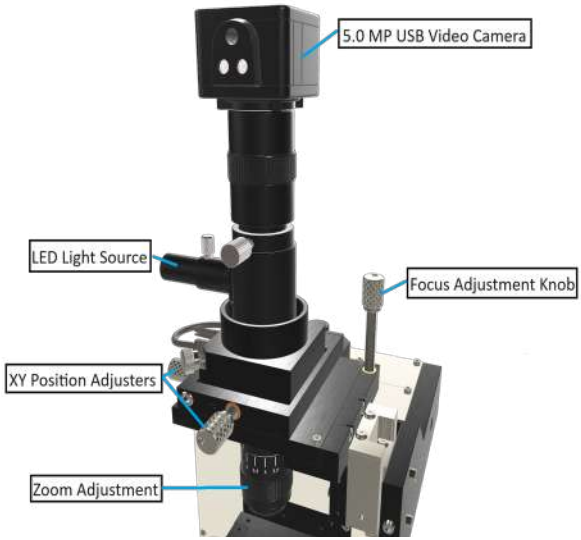
4. Adjust the height of the zoom tube so that the bottom of the tube is about 115 mm from the top of the AFM sample holder. Move the stage by hand and adjust the optical focus.

5. Place a probe in the probe holder (see section 4.3.) and place it into the light lever.

6. With the Z motor control, move the probe such that is about ½ in. above the sample plate.

7. With the focus adjustment knob, raise the zoom tube approximately ½ in.

8. Slowly move the stage by hand until the probe is viewed in the video microscope window. As required, adjust the focus, the lighting, and the position of the stage until the probe is in the center of the optical view




## A.6. First Alignment of the Light Lever

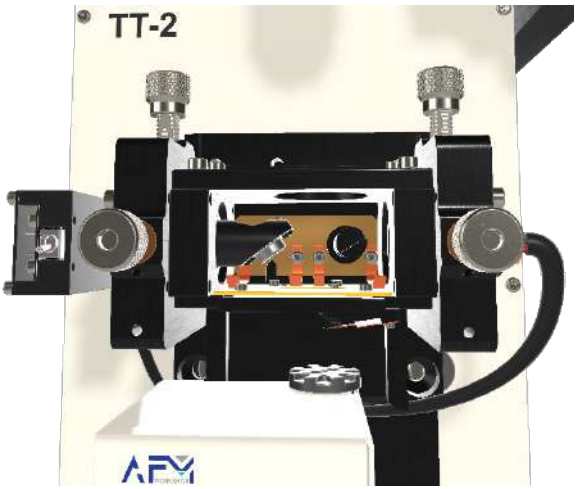
This procedure assumes that the light lever was correctly aligned at the factory or by an authorized agent.

1. Place the sample disk on the sample holder.

2. Place the probe holder in the light lever with a contact or vibrating mode probe installed.

3. Adjust the optical microscope until the probe/substrate can be viewed on screen.

4. Center the Photoluminescent Diode (PLD) and detector translation mechanisms with thumb screws

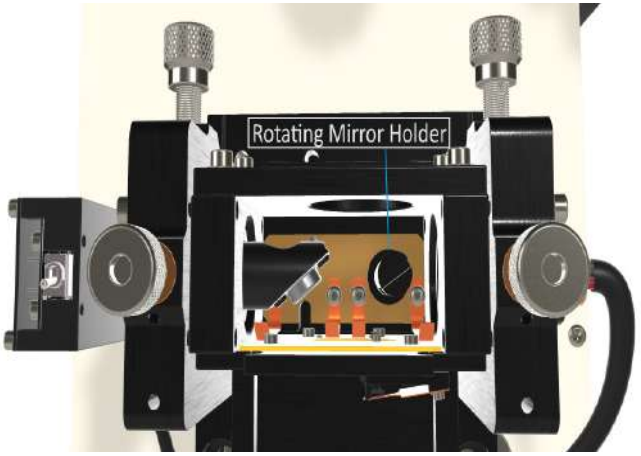



5. Turn on the Photoluminescent Diode (PLD).

6. Use a flathead screwdriver to rotate the mirror until the Photoluminescent Diode (PLD) spot is on the back side of the cantilever.

7. Lower the Z motor until the probe is just above the sample disk.

8. Rotate the mirror to move the Photoluminescent Diode (PLD) light until it is visible on the probe.

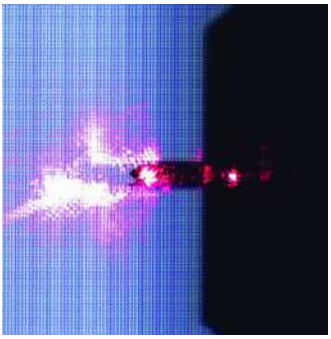


9. Loosen the set screws that hold the Photoluminescent Diode (PLD).

10. Slide the Photoluminescent Diode (PLD) in or out until it is focused.

11. Tighten the set screws to hold the Photoluminescent Diode (PLD) in place.

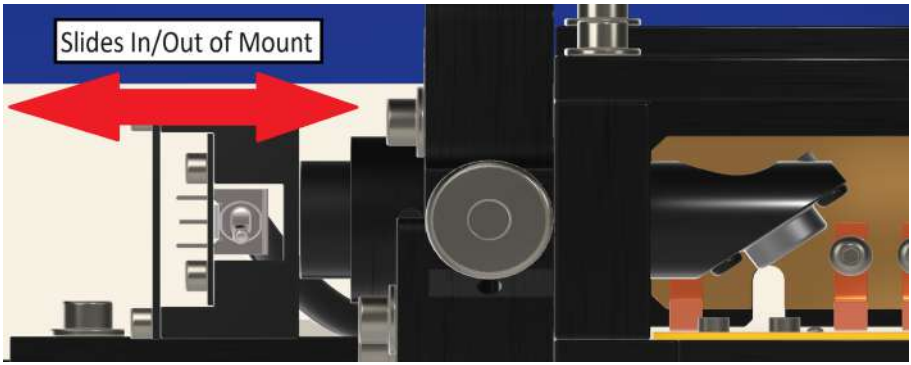
12. Move the Photoluminescent Diode (PLD) spot out to the edge of the cantilever.



13. Release the set screws that hold the photo-detector in place.

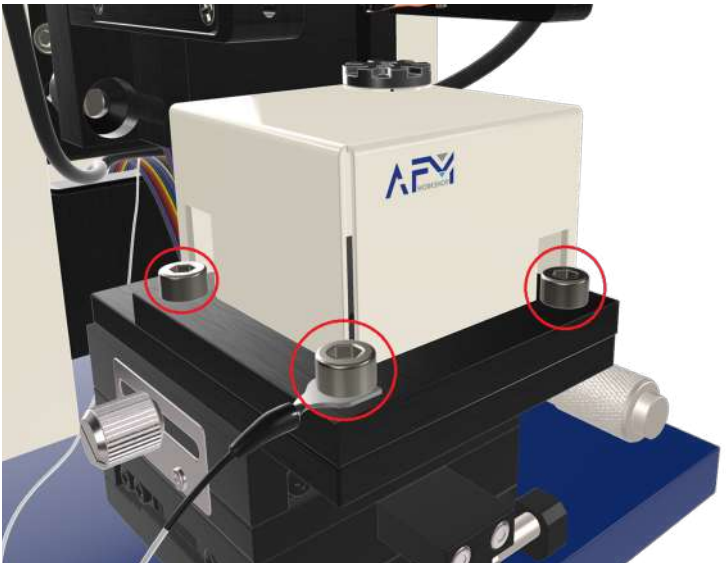
14. Move the photo-detector in and out until the T-B signal is zero.  Tighten the set screws.

15. Move the L-R signal to zero by adjusting the detector translation thumb screws.



## A.7. Installing the XYZ Piezoelectric Scanner

1. If not already turned off, turn off the Ebox electronics.



2. Place the scanner on the TT-2-AFM XY translator and locate the ribbon cable that plugs into the scanner.

3. Plug in the ribbon cable.  The connector is located on the back of the scanner.

4. Screw in the three socket screws that hold the scanner onto the XY translator.



**CAUTION: Do not twist the ribbon cable. Doing so will damage the TT-2-AFM electronics.**

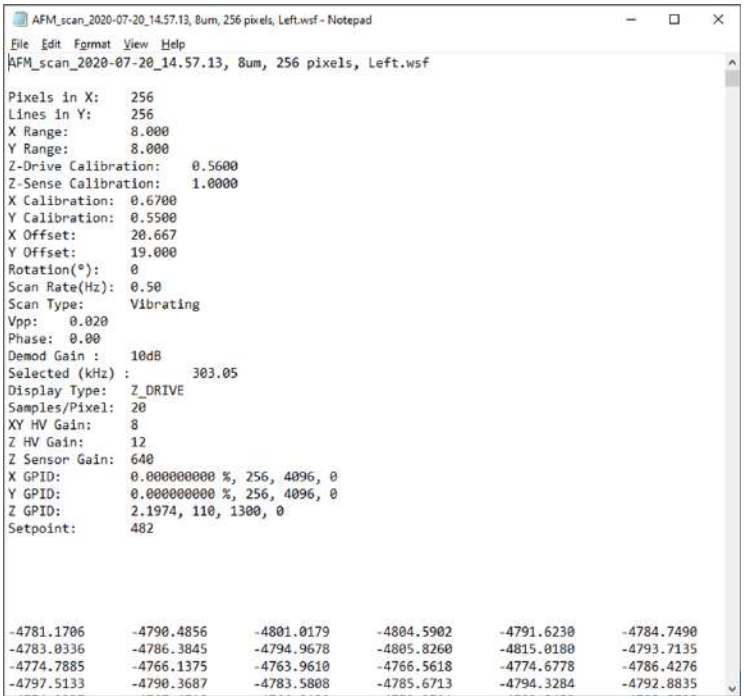
### A8. Grounding the TT-2-AFM

There are several grounding options for the TT-2-AFM. The green ground wire should be used to connect the stage with the Ebox. The probe is grounded by a grounding connector, which should be attached to the Modes/IO connection on the back of the stage. A wire can be added to the ground connector to receive signals from the probe.



## Appendix B: TT-2-AFM Data Files

TT-2-AFM data files are stored in a text format. At the beginning of the data file are stored many of the parameters used for the scan. It can be useful to check these in order to reproduce scanning conditions. The parameters can be easily observed by opening the file in a text editor, such as notepad, or using the “metadata Browser” function in Gwyddion.


AFM_scan_2020-07-20_14.57.13, 8um, 256 pixels, Left.wsf - Notepad

File Edit Format View Help

```
AFM_scan_2020-07-20_14.57.13, 8um, 256 pixels, Left.wsf

Pixels in X:    256
Lines in Y:     256
X Range:        8.000
Y Range:        8.000
Z-Drive Calibration:    0.5600
Z-Sense Calibration:    1.0000
X Calibration:  0.6700
Y Calibration:  0.5500
X Offset:       20.667
Y Offset:       19.000
Rotation(°):    0
Scan Rate(Hz):  0.50
Scan Type:      Vibrating
Vpp:    0.020
Phase:  0.00
Demod Gain :    10dB
Selected (kHz) :        303.05
Display Type:   Z_DRIVE
Samples/Pixel:  20
XY HV Gain:     8
Z HV Gain:      12
Z Sensor Gain:  640
X GPID:         0.000000000 %, 256, 4096, 0
Y GPID:         0.000000000 %, 256, 4096, 0
Z GPID:         2.1974, 110, 1300, 0
Setpoint:       482




-4781.1706      -4790.4856      -4801.0179      -4804.5902      -4791.6230      -4784.7490
-4783.0336      -4786.3845      -4794.9678      -4805.8260      -4815.0180      -4793.7135
-4774.7885      -4766.1375      -4763.9610      -4766.5618      -4774.6778      -4786.4276
-4797.5133      -4790.3687      -4783.5808      -4785.6713      -4794.3284      -4792.8835
```

# Appendix C: Calibration

Calibration of the SA AFM requires scanning a standard sample that has known X, Y and Z dimensions. If a sample with squares is used, the pitch in X and Y should be around 10 µm. The height of the features in Z should be around 100 nm.

1. Scan a standard sample with known dimensions in X, Y and Z at 0°. It is helpful to ensure that the sample is inserted with the features as straight as possible, to ensure accurate X and Y measurements.

2. Open the AFM Calibration software. Make sure to allow blocked content.

Welcome to the AFM Workshop calibration form.
Intended to be used with Internet Explorer

| Piezo | System Cal | Measures1 | Measures2 | Measures3 | Measures4 | Actual |
|---|---|---|---|---|---|---|
| X | | | | | | |
| Y | | | | | | |
| Z | | | | | | |
| | | | | | | Set XY |
| | | | | | | |

Calculate | Clear X | Clear Y | Clear Z | Reset

New X:
New Y:
New Z:

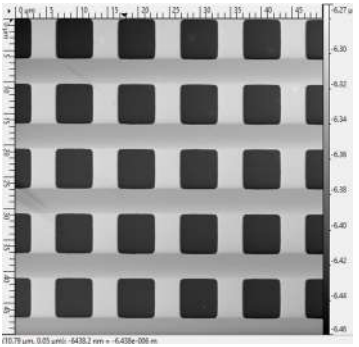
Update Calibrations ☐

3. Insert the current calibration values, located on the System tab, under System Cal in the calibration software.

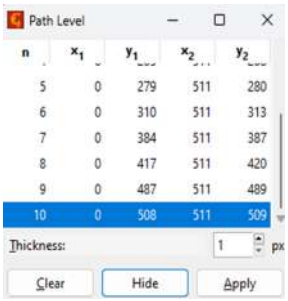
4. Insert the known values for the calibration standard in the Actual column in the calibration software. For the BudgetSensors HS-100MG, the outer sections have a 10 micron XY pitch and the inner sections have a 5 micron XY pitch. For the Z height, this is listed on the casing of the sample.

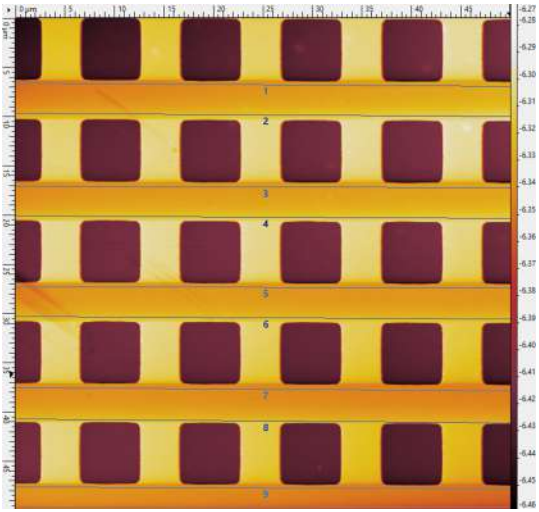
5. To calibrate, open the completed scan in Gwyddion and level the image using the following method. First, perform a 'Level Data by Mean Plane Subtraction.'



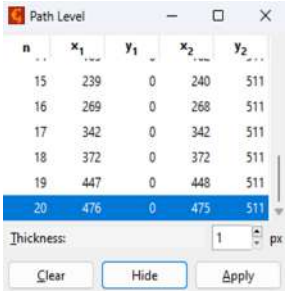



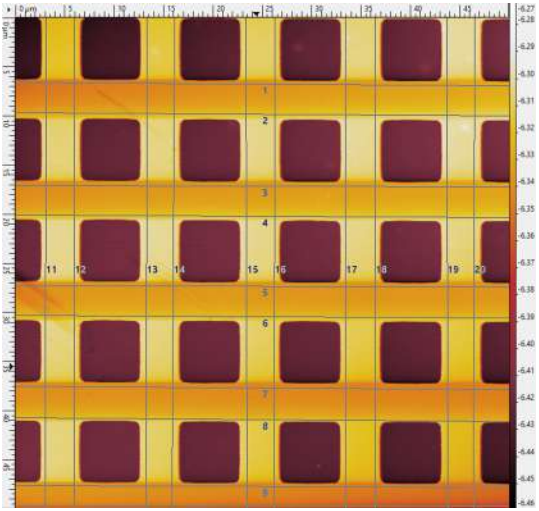
6. After this, navigate to the ‘Tools’ section and select ‘Level rows using intersections with given lines.’ Using this tool, draw horizontal lines across the scan from left to right, leaving space away from the features as shown. You may also need to add a top and/or bottom boundary if there is open space. As shown, 10 lines were required.



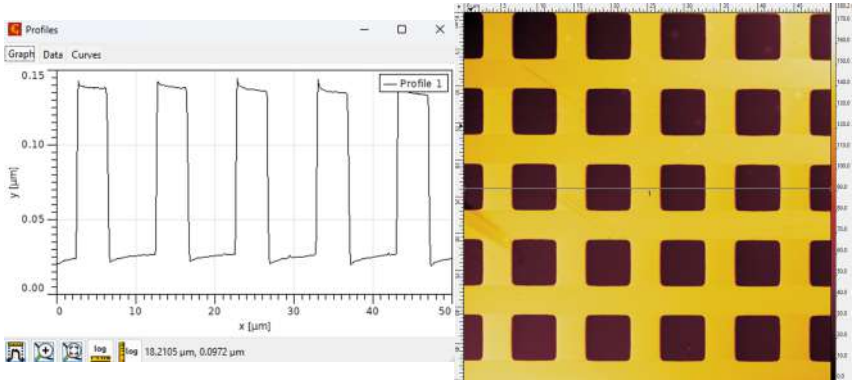
7. After drawing these horizontal boundaries, you will need to draw the vertical boundaries. Repeat the same process as before, going from the top of the scan to the bottom, leaving space between the features.



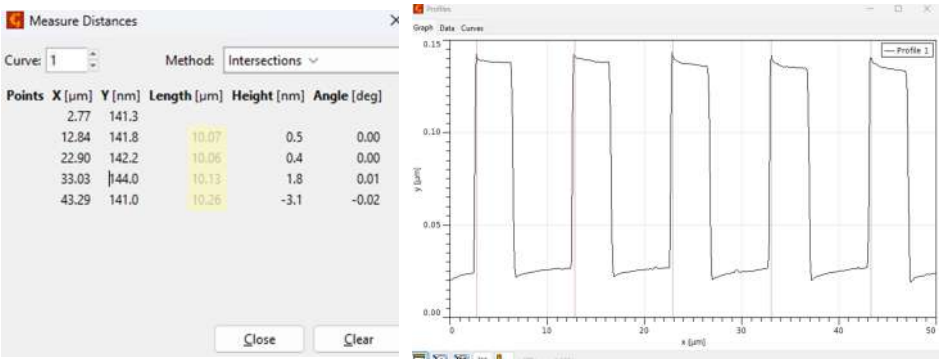



8. Once finished, click apply. The background should look uniform throughout. If not, press ‘CTRL + Z’ to undo and readjust the lines. Giving more space between the lines and features can help ensure a uniform background. Once satisfied with the leveling, click ‘Clear.’

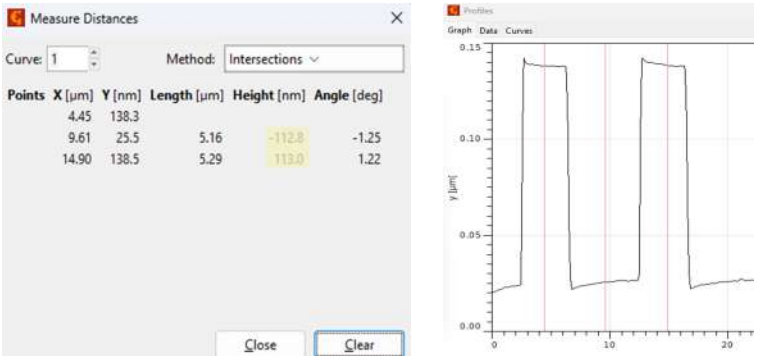
9. Once leveled, navigate to the ‘Tools’ section and select ‘Extract Profiles along arbitrary lines’. Once opened, left-click and drag to draw across the scan in the X direction across the features, as shown below. Once satisfied, click ‘Apply’ and the profile will pop up.




10. Once you have the profile window, click the icon in the bottom left labeled ‘Measure distances in graph’. Once open, measure the pitch from feature to feature as shown (beginning to beginning, end to end, etc.).




11. It is recommended to take 2 Z measurements from X and 2 Z measurements from Y. In order to do this, simply measure from the top of two features to the bottom of two features, as shown below. Then, input this data into the AFM Calibration software.



12. Using these X length and Z measurements, input them into the AFM Calibration software.

Welcome to the AFM Workshop calibration form.
Intended to be used with Internet Explorer

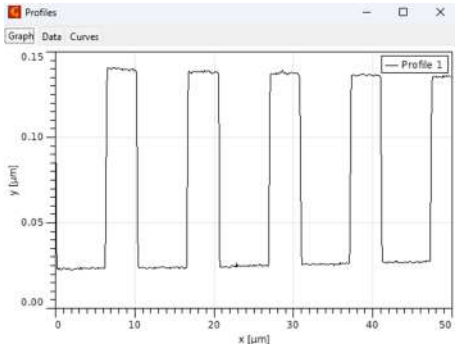
| Piezo | System Cal | Measures1 | Measures2 | Measures3 | Measures4 | Actual |
|---|---|---|---|---|---|---|
| X | 0.55 | 10.07 | 10.06 | 10.13 | 10.26 | 10 |
| Y | 0.55 | | | | | 10 |
| Z | 0.8 | 112.8 | 113.0 | | | 105 |
| | | | | | | Set XY |
| | | | | | | |

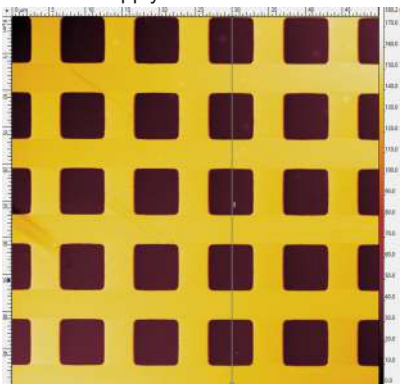
Calculate | Clear X | Clear Y | Clear Z | Reset

New X:
New Y:
New Z:

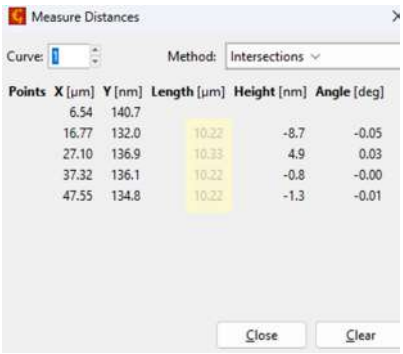
Update Calibrations ☐

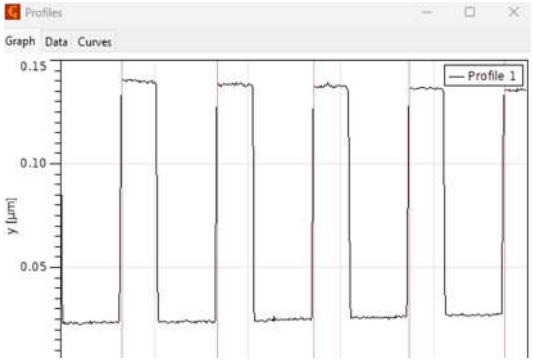
13. Now, follow the same process for taking the Y measurements and last two Z measurements. Begin by drawing the line profile vertically, then click 'Apply.'



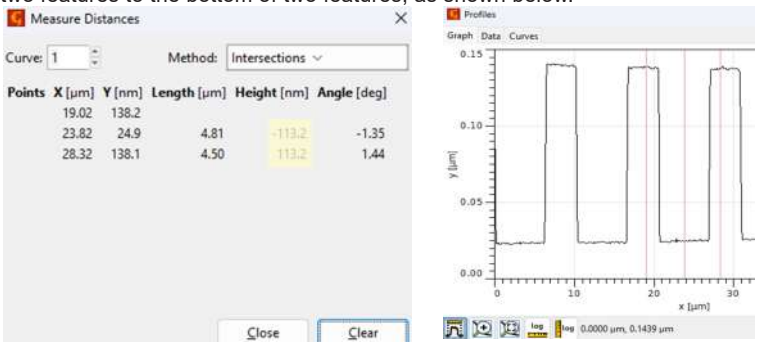



14. Click on 'Measure Distances in graph' and measure the pitch from feature to feature as shown (beginning to beginning, end to end, etc).



15. Then, proceed to take the final 2 Z measurements from the profile, measuring from the top of two features to the bottom of two features, as shown below.




16. Now, input the remaining data into the AFM Calibration software. It should look something like this.

Welcome to the AFM Workshop calibration form.
Intended to be used with Internet Explorer

| Piezo | System Cal | Measures1 | Measures2 | Measures3 | Measures4 | Actual |
|---|---|---|---|---|---|---|
| X | 0.55 | 10.07 | 10.06 | 10.13 | 10.26 | 10 |
| Y | 0.55 | 10.22 | 10.33 | 10.22 | 10.22 | 10 |
| Z | 0.8 | 112.8 | 113.0 | 113.2 | 113.2 | 105 |
| | | | | | | Set XY |
| | | | | | | |

Calculate | Clear X | Clear Y | Clear Z | Reset

New X:
New Y:
New Z:

Update Calibrations ☐

**17. Once the software has been fully filled out, click the 'Calculate' button. Your New X,Y, and Z calibration values will be shown below. From here you can use CTRL + C to copy these values, then paste them into the AFM Control Software system tab using CTRL + V. The values will automatically round to 3 decimal places - this is normal.**

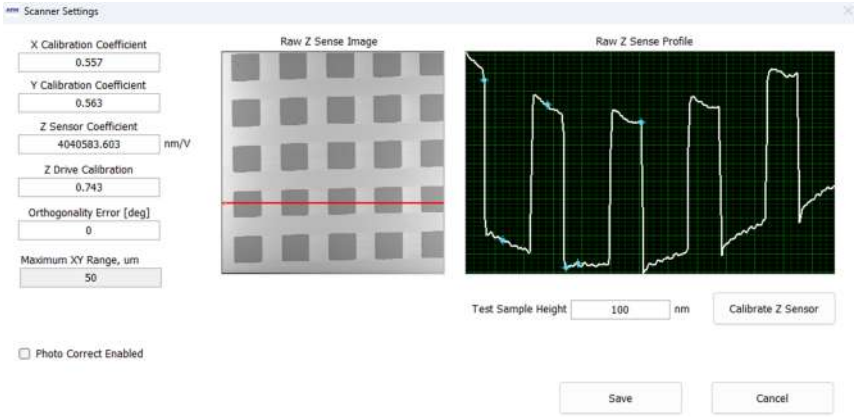
Welcome to the AFM Workshop calibration form.
Intended to be used with Internet Explorer

| Piezo | System Cal | Measures1 | Measures2 | Measures3 | Measures4 | Actual |
|---|---|---|---|---|---|---|
| X | 0.55 | 10.07 | 10.06 | 10.13 | 10.26 | 10 |
| Y | 0.55 | 10.22 | 10.33 | 10.22 | 10.22 | 10 |
| Z | 0.8 | 112.8 | 113.0 | 113.2 | 113.2 | 105 |
| | | | | | | Set XY |
| | | | | | | |

Calculate | Clear X | Clear Y | Clear Z | Reset

New X: 0.5570582428430406
New Y: 0.5632837277384729
New Z: 0.7430340557275542

Update Calibrations ☐




**NOTE: To obtain the most accurate calibration values for the scanner, the user may have to complete multiple calibration iterations. Additionally, the calibration should be done using a sample of similar height to your features of interest (e.g. 2D Material scanning requires Z calibration with an atomic terrace sample, like HOPG or SiC. Calibration is also required at each Z HV gain, found in the resolution settings. It's recommended to create new scanner profiles or keep a notebook with these Z calibration values writen down for reference. This will maximize the AFM's accuracy in measurements.**

### C.1. Calibrating Z Sensor

For any SA AFM with a 17 micron Z piezo, calibrating the Z Sensor is also an important step in calibration. Unlike Calibrating the X, Y, and Z Drive values, the Z Sensor can be done entirely through the AFMWorkshop Control Software. Z Sense is used for a variety of optional modes and scan types, including Z Sense Scanning, Lithography, and Advanced/ Basic Force Distance measurements.

1. First, start by opening the software and preparing to take a normal calibration scan as shown before. The sample should be placed as straight as possible. Go through the Pre-Scan tab steps, then navigate to the Topo Scan window.

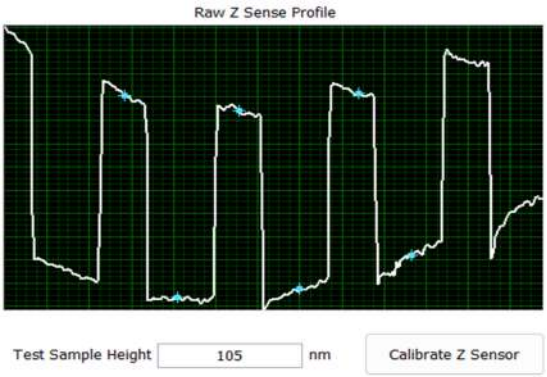
2. Once you are on the Topo Scan window, select 'Z Sense' as your 'Left Image' and 'Z Drive' as your right image. Begin scanning and adjust your Z GPID if necessary to ensure a quality image is obtained.

3. After the scan has completed, click 'Save' and navigate back to the Pre-Scan tab.

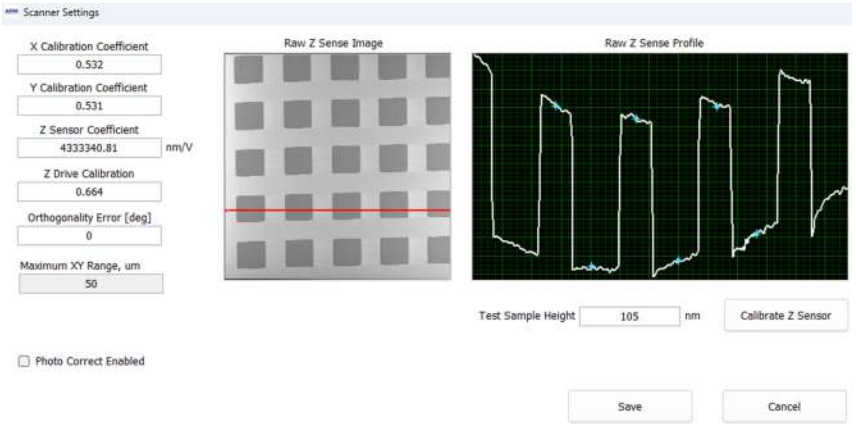
4. Open up the 'System Settings' using the gear under the 'System' section of the Pre-Scan tab.

5. From here, you'll notice the 'Raw Z Sense Image' and 'Raw Z Sense Profile' windows. Drag the horizontal red line across your feature set until you are satisfied with the line profile.

6. Once you've set the line across the features, you'll notice small blue dots along the line profile. These dots can be clicked and dragged to position. You should drag these blue dots until you have an alternating pattern of Top of Feature - Bottom of Feature - Top of Feature - etc, then set the 'Test Sample Height' to the height listed on the BudgetSensors HS-100MG Sample case. See below for reference.



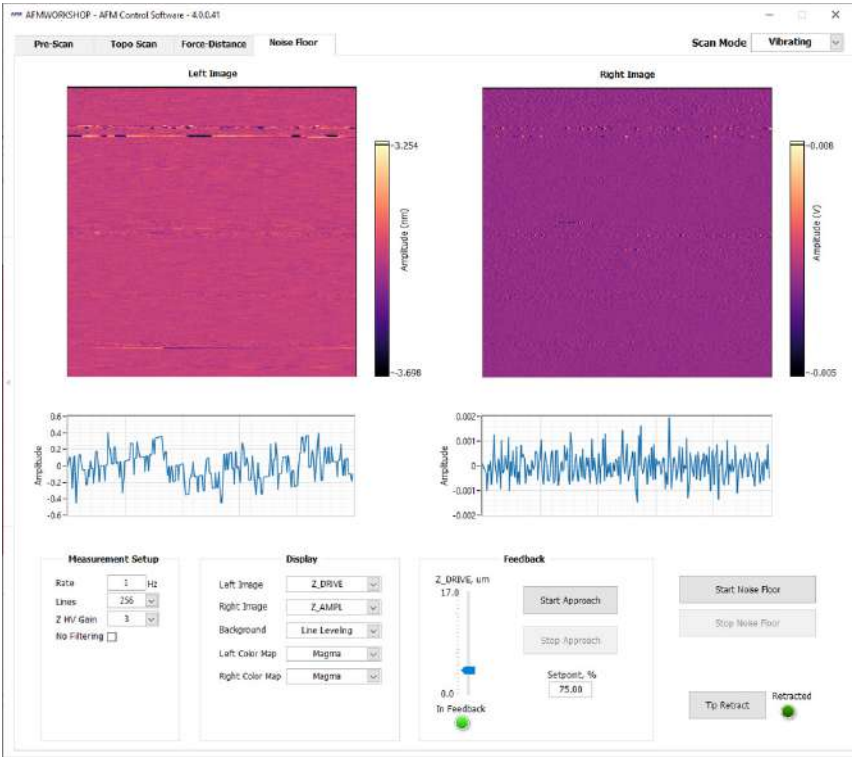

**7. Once you've set these blue dots and the test sample height in the software, click the 'Calibrate Z Sensor' button. This will automatically update your Z Sensor calibration value. Finally, click save to save the calibration changes in the software.**




## Appendix D: Noise Floor

To measure the noise floor, it is necessary to scan the sample with no voltage applied to the X and Y piezoelectric ceramics. The AFM Control Software will then measure the structural noise of the system when the tip is in feedback with the surface. There is a special routine built into the AFM Control Software for this purpose Make sure the tip is a sizeable distance from the surface.  Initiate the Tip Retract function, if necessary.

1. Press the key combination CTRL-N to activate the Noise Floor measurement tab, as shown.




2. In the noise floor tab, set the scan parameters for noise measurements:

- Scan Rate = 0.5 Hz
- Scan Lines = 256
- Left, Right Image = Z-Drive, Z-Error
- Background: Line Leveling (optional)

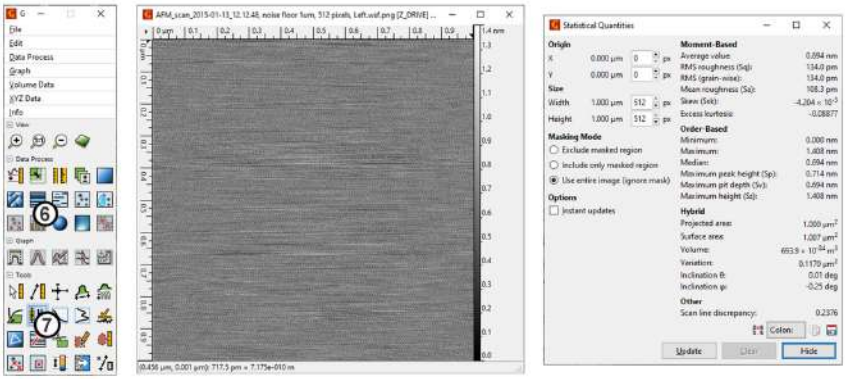
3. Initiate Automated Tip Approach.

4. Begin a scan.

5. To analyze the noise floor measurement, open the completed Z-Drive scan in Gwyddion.

6. Level the image by pressing the Polynomial level button, 1st order.

7. Find the noise value with the Statistical Quantities button, under Ra (Sa) for surface roughness.



Noise can be reduced by lowering the Z HV Gain using the Z HV Gain setting in the Noise floor tab. This should be done with the probe out of contact with the surface. When the Z gain is lowered, a new tip approach must be initiated because the Z ceramic's dynamic range has been reduced.



**CAUTION: Increasing the Z HV Gain will increase the Z ceramic's dynamic range, and will raise the sample stage. Make sure the tip is removed from the surface before increasing Z HV Gain.**

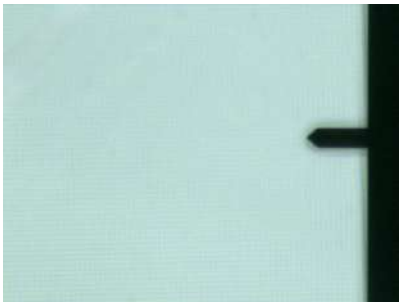
## Appendix E: Probes

There are two major types of probes, vibrating mode and non-vibrating mode probes. Each of these probes may have a metallic coating, which makes them more reflective.

**1. Vibrating Mode Probes**

Vibrating mode probes are in the shape of a rectangle. Typical VM probe specifications:

| | |
|---|---|
| Resonance Frequency | 150 – 350 kHz |
| Length | 125 – 225 µm |
| Width | 30 – 40 µm |
| Thickness | 0.5 – 1 µm |
| Force Constant | 15 – 50 N/m |

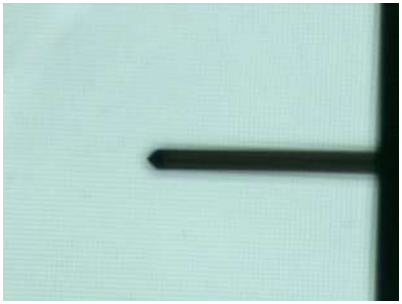


ACLA probes from Applied Nanostructures are recommended.

**2. Non-Vibrating mode (contact) Probes**

Non-vibrating mode probes can be in the shape of a V or rectangular. Typical Non-Vibrating mode probe specifications are:

| | |
|---|---|
| Resonance Frequency | 7 – 200 kHz |
| Length | 225 – 450 µm |
| Width | 30 – 40 µm |
| Thickness | 0.1 – 0.5 µm |
| Force Constant | 0.1 – 0.5 N/m |

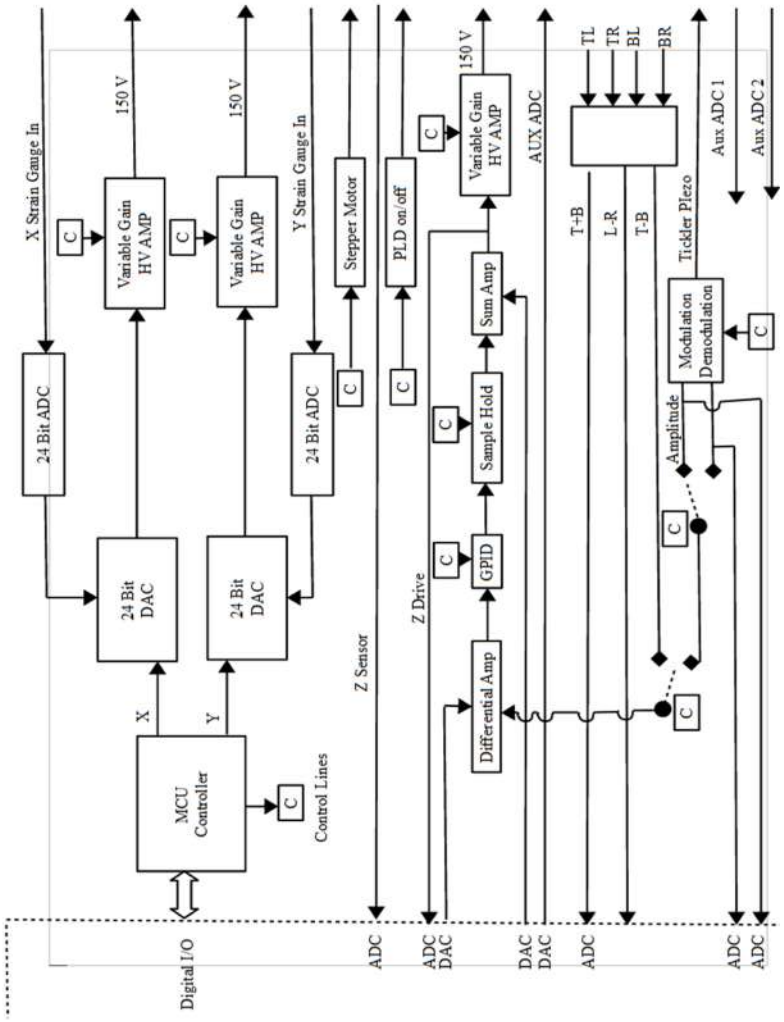


SHOCONA probes from Applied Nanostructures are recommended.

# Appendix F: Technical Information

This appendix has technical information that may be useful for creating customer instrumentation with the TT-2-AFM. Additional information is available in the TT-2-AFM technical guide.

## F.1. Electronic Block Diagrams



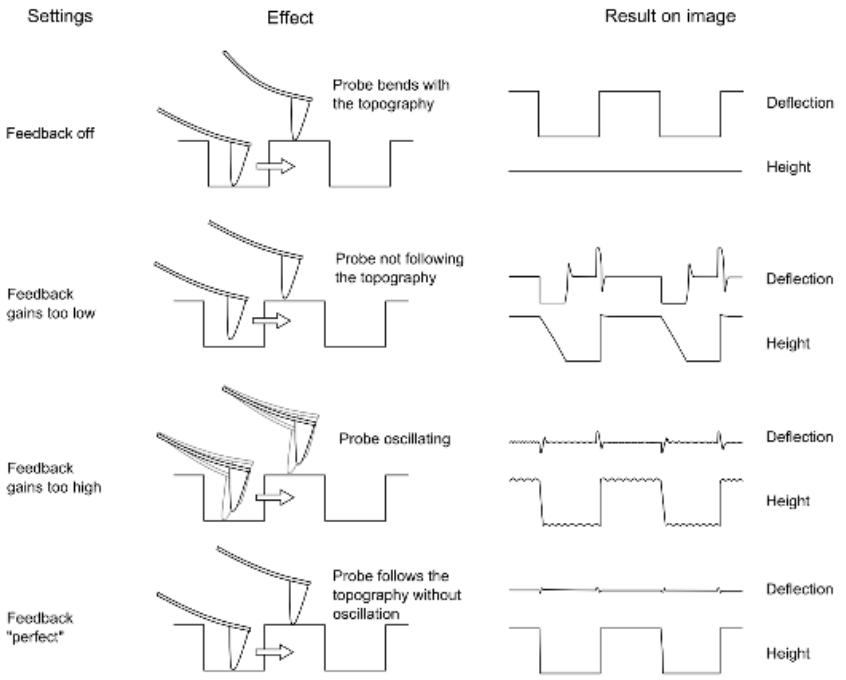
50 Pin Connector Assignment

| Pin No. | Function | Pin No | |
|---|---|---|---|
| 1 | X Sense | 2 | X Ref |
| 3 | X Drive | 4 | Ground |
| 5 | Y Sense | 6 | Y Ref |
| 7 | Y Drive | 8 | Ground |
| 9 | Z Sense | 10 | Z Ref |
| 11 | Z Drive | 12 | Ground |
| 13 | Z Err | 14 | Ground |
| 15 | Z Set | 16 | Ground |
| 17 | E 125 | 18 | Ground |
| 19 | Z Amp | 20 | Ground |
| 21 | Z Phase | 22 | Ground |
| 23 | T-B | 24 | Ground |
| 25 | L-R | 26 | Ground |
| 27 | T+B | 28 | Ground |
| 29 | Z DAC | 30 | Ground |
| 31 | Aux ADC 1 | 32 | Ground |
| 33 | Aux ADC 2 | 34 | Ground |
| 35 | Tickler | 36 | Ground |
| 37 | Ref AC + | 38 | Ground |
| 39 | Ref AC - | 40 | Ground |
| 41 | E71 | 42 | Ground |
| 43 | Laser Power | 44 | Ground |
| 45 | +12 V | 46 | Ground |
| 47 | -12 V | 48 | Ground |
| 49 | +5 V | 50 | Ground |

# Appendix G: Optimizing GPID Parameters

The Z feedback scan parameters, such as the setpoint voltage and the GPID parameters are adjusted as the line scan is being made. The goal for adjusting these parameters is to have the probe perfectly track the surface. The probe is tracking the surface best when the Z error image has a minimal signal. Establishing the optimal conditions requires practice and care.

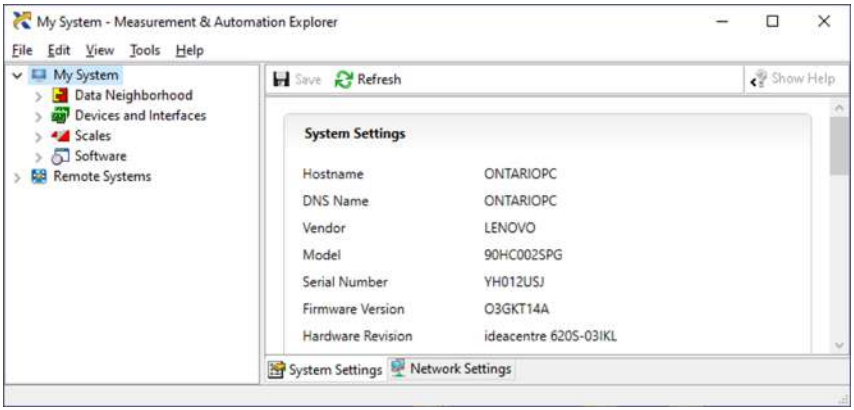
When first learning to operate an AFM, it is helpful to scan a reference sample in 2D and adjust the GPID settings to see the effect on the Z voltage and the Z error signal, as shown in the figure below. The Option for 2D imaging is located on the topo scan tab, and will continuously scan the same image line over and over, allowing the user to easily see the effect of adjusting GPID and setpoint parameters. The image below schematically shows the effect of GPID parameter s that are too low or too high.



# Appendix H: Renaming National Instruments DAQ Device

The following procedure will allow the user to reset the National Instruments DAQ card registry with the device number corresponding with the user's Ebox. This may be useful if the user's AFM Control Software does not initialize. This unusual occurrence can occur if multiple version of the software have been installed.

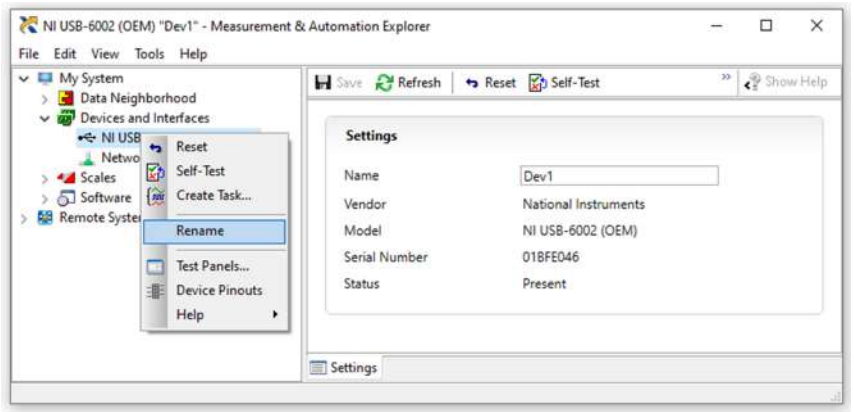
Open the NI MAX:  Measurement and Automation Explorer software as follows:




1. Expand Devices and Interfaces under My System on the left scroll tab.

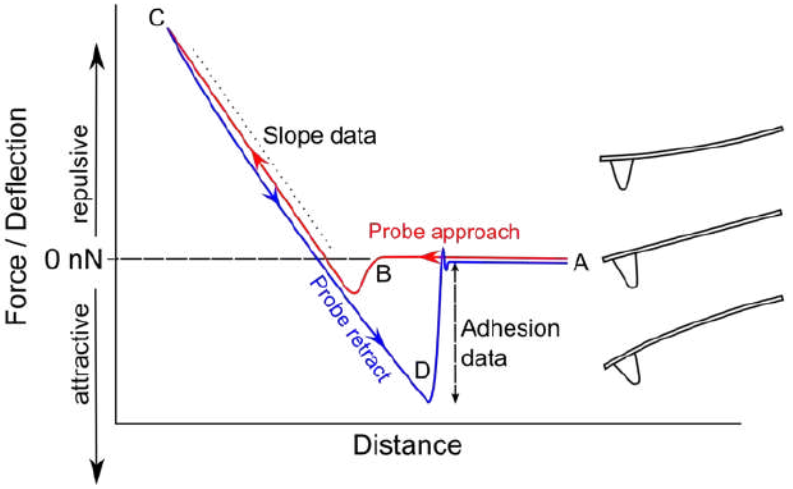
2. Making sure the correct Ebox USB is plugged into the computer, delete any devices that are not green. There should only be one green device, corresponding with the DAQ card in the user's Ebox.

3. Rename the only remaining green device as Dev1. This can be accomplished by right-clicking on the device and selecting Rename.

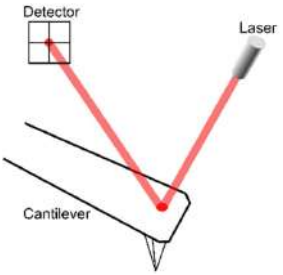


# Appendix I: Analyzing Force/Distance Curves

A force/distance curve is a graph that shows the amount a cantilever tip bends as it is systematically pushed into a sample. The probe may be lowered onto the surface, or the surface may be raised into the probe.




What is actually measured is the deflection of the cantilever, or the Top - Bottom (T-B) signal of the photo-detector.




The deflection of the cantilever can be converted to Force - usually expressed In NanoNewtons (nN) - using Hooke's law:

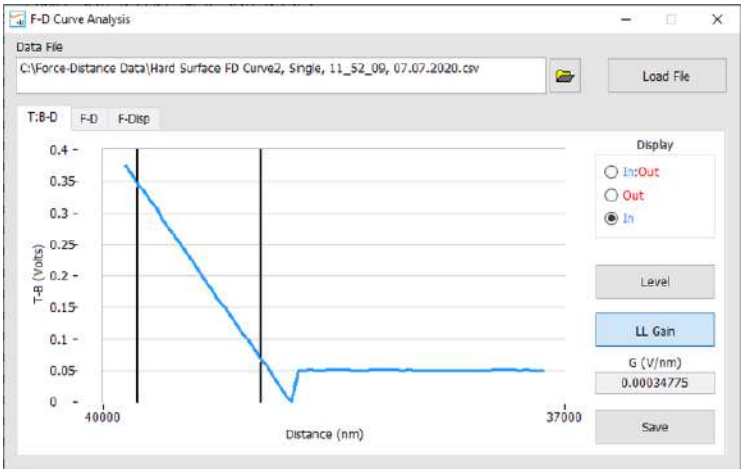
Force = - K × d

K = force constant (nN/nm)

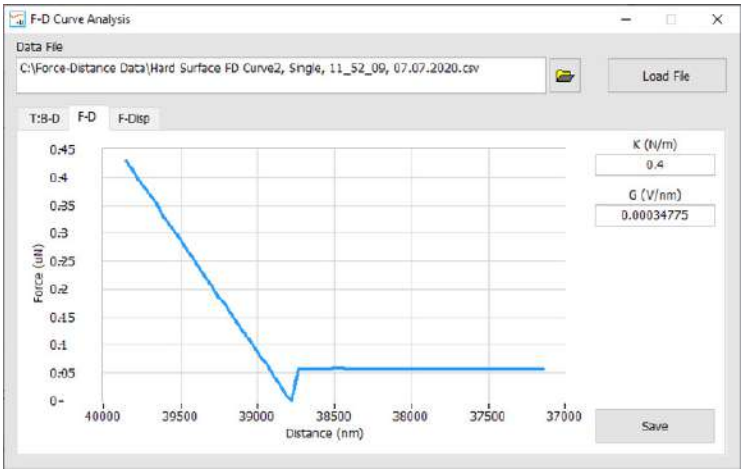
d = deflection of the cantilever (nm)

In a light lever AFM, the user must convert the T-B signal into the cantilever deflection. This is achieved by measuring the change in the T-B signal as the sample is directly pushing against the probe. This is labeled slope data in the first figure above. The force constant for the cantilever is supplied by the manufacturer.

A force/distance curve from the TT-2-AFM is illustrated in the figure below. The light lever gain can be determined simply in any spreadsheet application. However, the AFMWorkshop F-D Curve Analysis applet was used below.




The data above shows the Deflection vs. distance curve, which can be used to calculate the light lever gain. This was calculated using the region between the two black lines. As shown this was found to be 0.3 mV /nm. The second image below uses this value, along with the value of spring constant (K) to convert the data into a force-distance curve.



The final image below shows the curve converted to Displacement-distance curve. In the case where no indentation of compression of the sample occurs, as shown here, the curve shows a vertical line upon contact with the surface. This indicates that the probe was not moving at this point, and the only change was bending of the probe. In experiments such as nanoindentation, there will be a slope in this region, However, a curve as shown here is typically used to calibrate the system during each experimental run.

